The SRDS Formal Framework

A Self-Contained Mathematical Theory of Critical Transitions
in Self-Referential Dissipative Systems

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Abstract

A system that accumulates damage under load until a characteristic yield point, past which return is improbable, recurs across physically disjoint substrates. We develop, in full and self-contained form, the mathematical class behind this pattern: *self-referential dissipative systems* (SRDS), sequences in which the state recursion is driven by an update operator that is itself rewritten by the state it produces. The class is fixed by five axioms—self-reference, dissipation, accumulation, transition, irreversibility—together with four regularity conditions. Only self-reference and dissipation are primitive: under the explicitly named *constitutive* conditions L1–L4, accumulation, transition, and irreversibility follow as *theorems* rather than independent postulates. The load-bearing assumption is the saturation law L2—a single-peaked, state-dependent damage rate, the operative form of self-reference—from which the mean cumulative curve inherits its sigmoid inflection *in expectation*, after the manner of a phenomenological constitutive law; the pathwise specimen inflection is the fitted empirical object, and the conditional form of irreversibility rests on the elastic-below-yield return protocol L4. The reduction is thus a consistency result under named postulates, not a derivation from self-reference alone.

On this base we prove three structural results. *T1*, an information–stability correspondence, ties the sign of the conditional phase-space volume rate—the sum of conditional Lyapunov exponents, equivalently minus the conditional dissipation rate—to the transient slope of mutual information, with the maximal-exponent reading as the dominant-mode case. *T2*, an autopoietic stationarity theorem, yields under a contraction hypothesis a unique, globally attracting non-equilibrium steady state of the operator recursion. *T3*, a Goldilocks principle, places optimal function in a neighbourhood of criticality rather than exactly at it. We then construct the two-scale renormalization operator that carries the dynamics across observational scales and establish the existence and uniqueness of its monotone fixed point—Schauder existence, sup-norm Banach uniqueness in the gap/damping regime, and Birkhoff–Hopf projective contraction governing the smoothing channel's synchronization rate—identifying the cumulative inflection as the order parameter of the class. A concluding No-Go theorem shows that no universal numerical inflection location exists within reparametrization-closed operator classes: the symmetric value is recovered only under mirror symmetry with fixed-point uniqueness and a non-degenerate mode, and is otherwise a substrate-specific measured quantity.

We close with an elimination ledger—making precise what cannot be concluded without additional structure—and with the spectral and contraction substructure that governs convergence rates. Every result is proved in full, reduced to a cited result of the established literature, or stated under explicitly named hypotheses, with the conditions on which it rests made explicit. The framework is metric-agnostic: it is instantiated by, but does not depend on, any particular measurement system.

Contents

1 Preliminaries and Notation

This section fixes the analytic and information-theoretic vocabulary used throughout and records the standing regularity assumptions. All statements are proved or reduced to standard references.

1.1 Function spaces

Let $C([0,1])$ denote the space of continuous real functions on $[0,1]$ with the supremum norm $\|Q\|_\infty := \sup_{x\in[0,1]} |Q(x)|$.

Proposition 1.1 (Completeness)**.** *$(C([0,1]), \|\cdot\|_\infty)$ is a complete metric space.*

Proof. Let $(Q_n)_{n\in\mathbb{N}}$ be Cauchy in $\|\cdot\|_\infty$. For each fixed x, $|Q_n(x) - Q_m(x)| \le \|Q_n - Q_m\|_\infty$, so $(Q_n(x))_n$ is Cauchy in $\mathbb{R}$ and, by completeness of $\mathbb{R}$, converges to a limit we call $Q(x)$. The convergence is uniform: given $\varepsilon > 0$ choose N with $\|Q_n - Q_m\|_\infty < \varepsilon$ for $n, m \ge N$; letting $m \to \infty$ in $|Q_n(x) - Q_m(x)| < \varepsilon$ gives $|Q_n(x) - Q(x)| \le \varepsilon$ for all x and all $n \ge N$, i.e. $\|Q_n - Q\|_\infty \le \varepsilon$. A uniform limit of continuous functions is continuous: for $n \ge N$ and $|x - x_0|$ small enough that $|Q_n(x) - Q_n(x_0)| < \varepsilon$ (continuity of Q_n), the triangle inequality $|Q(x) - Q(x_0)| \le |Q(x) - Q_n(x)| + |Q_n(x) - Q_n(x_0)| + |Q_n(x_0) - Q(x_0)| < 3\varepsilon$ holds. Hence $Q \in C([0,1])$ and $Q_n \to Q$. □

Let $BV([0,1])$ denote the functions of bounded variation, with total variation

$$\mathrm{Var}(Q) := \sup_P \sum_i |Q(x_i) - Q(x_{i-1})|$$

over partitions $P = \{0 = x_0 < \cdots < x_k = 1\}$.

Proposition 1.2. *Every $Q \in BV([0,1])$ is bounded, with $|Q(x)| \le |Q(0)| + \mathrm{Var}(Q)$ for all x.*

Proof. For fixed x the partition $\{0, x\}$ gives $|Q(x) - Q(0)| \le \mathrm{Var}(Q)$, whence $|Q(x)| \le |Q(0)| + |Q(x) - Q(0)| \le |Q(0)| + \mathrm{Var}(Q)$. □

We write $\mathcal{M} \subset BV([0,1])$ for the set of *anchored monotone survival profiles*: nonincreasing Q with $Q(0) = 1$ and $Q(1) = 0$. Such a Q is exactly the survival function $1 - F$ of a probability measure F on $[0,1]$ (the right anchor $Q(1) = 0$ is intrinsic to the survival reading, not optional); the cumulative-damage curve $s(t)$ of Section 2 is the complementary CDF $1 - Q$, with the *same* inflection location $\hat{a}$. The renormalization operator of Section 8 acts on a gap-bounded admissible subclass $\mathcal{U}_\rho \subset \mathcal{M}$ (Definition 8.3). By Proposition 1.2, $\mathcal{M}$ is uniformly bounded.

1.2 Information theory

Definition 1.3. For a discrete random variable X with law p, the Shannon entropy is $H(X) := -\sum_x p(x) \log p(x)$ (convention $0 \log 0 = 0$). For a pair (X, Y) the joint entropy is $H(X, Y) := -\sum_{x,y} p(x, y) \log p(x, y)$, the conditional entropy $H(X \mid Y) := H(X, Y) - H(Y)$, and the mutual information $I(X; Y) := H(X) + H(Y) - H(X, Y)$.

Lemma 1.4 (Properties of mutual information)**.** *I satisfies: (i) $I(X; Y) \ge 0$; (ii) $I(X; Y) = I(Y; X)$; (iii) $I(X; Y) = 0$ iff X, Y are independent; (iv) $I(X; Y) \le \min\{H(X), H(Y)\}$.*

Proof. Write $I(X; Y) = \sum_{x,y} p(x, y) \log \frac{p(x,y)}{p(x)p(y)}$. (i) By Jensen's inequality applied to the convex function $t \mapsto t \log t$, equivalently by the nonnegativity of the Kullback–Leibler divergence $D(p(x, y) \,\|\, p(x)p(y)) = I(X; Y) \ge 0$ [8]. (ii) Symmetry is immediate from $I = H(X) + H(Y) - H(X, Y)$. (iii) Equality $I = 0$ holds iff $D = 0$, i.e. $p(x, y) = p(x)p(y)$ everywhere, which is independence. (iv) $I(X; Y) = H(X) - H(X \mid Y)$ and $H(X \mid Y) \ge 0$ for discrete variables give $I \le H(X)$; symmetry gives $I \le H(Y)$. □

Theorem 1.5 (Data-processing inequality)**.** *If $X \to Y \to Z$ is a Markov chain then $I(X; Z) \le I(X; Y)$.*

Proof. Standard; see [8, Thm. 2.8.1]. Processing of Y cannot increase information about X. □

For continuous laws we use the differential entropy $h(X) := -\int p \log p$ and the Gaussian maximum-entropy bound: among densities on $\mathbb{R}^n$ with covariance Σ, the Gaussian maximizes h, with $h(X) \le \frac{1}{2} \log((2\pi e)^n \det \Sigma)$ [8, Thm. 8.6.5].

1.3 The measurement model

The framework is metric-agnostic: it constrains the *geometry of a trajectory*, not the instrument that records it. We fix the minimal measurement vocabulary.

Definition 1.6 (Specimen and deformation curve)**.** A *specimen* is a finite ordered sequence $(z_1, \ldots, z_T)$ of states $z_t \in \mathbb{R}^m$, obtained from an embedding map φ of the raw sequence into $\mathbb{R}^m$. The index t is ordered loading time, not an i.i.d. sample index; the ordered sequence is the *deformation curve* of the specimen. For window size W and $t \ge W$ the window is $\mathcal{W}(t) := \{t - W + 1, \ldots, t\}$ (with $\mathcal{W}(t) := \{1, \ldots, t\}$ for $t < W$).

Because the index is loading time, successive measurements along a deformation curve are *autocorrelated by construction*; this is signal, not nuisance, and inference at the specimen level must treat the whole curve as the unit of analysis (Section 4).

1.4 Standing regularity assumptions

The dynamical results invoke the following conditions. They are sufficient, not necessary, and each is satisfied by standard generative models of ordered embedded sequences.

Assumption R1 (Stochastic flow)**.** The state sequence (z_t) is generated by a discrete-time process whose continuous-time interpolation solves $dz = F(z)\,dt + \sigma\,dW_t$, with $F : \mathbb{R}^m \to \mathbb{R}^m$ locally Lipschitz, $\sigma > 0$, and W_t standard Brownian motion in $\mathbb{R}^m$.

Assumption R2 (Lyapunov spectrum)**.** Along trajectories the variational equation $\dot{\xi} = DF(z(t))\,\xi$ admits a maximal Lyapunov exponent $\lambda_{\max} := \lim_{t\to\infty} \frac{1}{t} \log \|\xi(t)\| \,/\, \|\xi(0)\|$, which exists for a.e. initial condition by the multiplicative ergodic theorem [22].

Assumption R3 (Bi-Lipschitz embedding)**.** On the support of the trajectory law, φ is bi-Lipschitz: there are $0 < c_1 \le c_2 < \infty$ with $c_1\, d_{\mathcal{X}}(s, s') \le \|\varphi(s) - \varphi(s')\| \le c_2\, d_{\mathcal{X}}(s, s')$ for a base metric $d_{\mathcal{X}}$.

Assumption R4 (Ergodicity)**.** (z_t) is stationary and ergodic with invariant law μ having log-concave density on $\mathbb{R}^m$; hence time averages converge to μ-averages [3] and concentration holds [5].

Assumption R5 (Finite information)**.** For every t and every finite measurable partition, the discretized state coordinates have finite Shannon entropy, so $I(\cdot;\cdot)$ is well defined.

2 The SRDS Axiomatic Class

2.1 Two recursions

Definition 2.1 (Self-referential system)**.** A self-referential dissipative system is a pair of coupled recursions

$$z_{t+1} = F_t(z_t, u_t), \qquad F_{t+1} = \Phi(F_t, z_t, u_t), \tag{1}$$

where $z_t \in \mathbb{R}^m$ is the state, u_t an exogenous input, $F_t \in \mathcal{F}$ the update operator drawn from a complete metric space $(\mathcal{F}, d_{\mathcal{F}})$, and $\Phi : \mathcal{F} \times \mathbb{R}^m \times \mathcal{U} \to \mathcal{F}$ the meta-operator. A configuration F^* is *autopoietic* if $\Phi(F^*, z^*, u^*) = F^*$.

The first recursion is classical feedback dynamics [31, 1]. It is insufficient alone: a specimen remembers its loading history (the Bauschinger effect), a sequence carries path dependence. The second recursion makes that memory explicit: the operator that writes the next state is itself rewritten during writing. This is the operative closure of [20].

2.2 The five axioms

We collect the cumulative dissipation along a trajectory. Under Assumption R1 let $V : \mathcal{F} \to [0, \infty)$ be a measurable functional and define the one-step expected decrement and the cumulative damage

$$\delta_t := V(F_t) - \mathbb{E}[V(F_{t+1}) \mid \mathcal{G}_t]\,, \qquad \Sigma_t := \sum_{s<t} \delta_s, \tag{2}$$

where $(\mathcal{G}_t)$ is the natural filtration.

Axiom A0 (Self-reference)**.** The update operator evolves by the second recursion of (1): $F_{t+1} = \Phi(F_t, z_t, u_t)$; the operator is itself a state, rewritten as it acts.

Axiom A1 (Dissipation)**.** There is a Lyapunov functional $V \geq 0$ with $\mathbb{E}[V(F_{t+1}) \mid \mathcal{G}_t] \leq V(F_t)$, i.e. $\delta_t \geq 0$ in conditional expectation. Equivalently the system exports entropy to maintain bounded internal organization far from equilibrium [24].

Axiom A2 (Accumulation)**.** The damage Σ_t is monotone nondecreasing: $\Sigma_{t+1} \geq \Sigma_t$.

Axiom A3 (Transition)**.** The normalized damage $\sigma(t) := \Sigma_t/\Sigma_\infty$ admits a unique inflection point $\hat{a}$ (sigmoid form), the order parameter of the trajectory [27].

Axiom A4 (Irreversibility)**.** A trajectory that has crossed the inflection ($\sigma > \hat{a}$ in the normalized damage $\sigma = \Sigma/\Sigma_\infty$, equivalently $\Sigma > \hat{a}\,\Sigma_\infty$) returns to its pre-transition configuration with probability strictly less than one [28], where the *return probability* is the protocol-conditioned reverse likelihood of the crossing ensemble (Regularity L4, part L4a; made precise in Theorem 2.2(iii)) — the object the fluctuation relation controls, not an ordinary pathwise return frequency.

2.3 Regularity conditions

Regularity Condition L1 (Lyapunov half-order)**.** V is bounded below (normalized $V \geq 0$) and adapted to $(\mathcal{G}_t)$, so the increments δ_t are nonnegative adapted random variables and Σ_t is a nondecreasing adapted process, with $\mathbb{E}[V(F_0)] < \infty$ and $0 < \mathbb{E}[\Sigma_\infty] < \infty$ (finite, nontrivial total damage; for pathwise normalized statements $\sigma = \Sigma/\Sigma_\infty$ we take $\mathbb{P}[\Sigma_\infty > 0] = 1$, zero-damage paths counted as non-crossing).

Regularity Condition L2 (Constitutive damage-rate law)**.** The mean cumulative damage $s(t) := \mathbb{E}[\Sigma_t]/\mathbb{E}[\Sigma_\infty] \in [0, 1]$ evolves by an *autonomous* law in which the dissipation rate is a function of the accumulated state, $\dot{s} = g(s)$, with $g : [0, 1] \to [0, \infty)$ continuous and C^1 on $(0, 1)$ and: *(a)* $g(s) > 0$ on $(0, 1)$ (damage accrues while unsaturated); *(b)* $g(1) = 0$ (saturation: the rate vanishes at the autopoietic configuration); *(c)* g is single-peaked — there is $s^* \in (0, 1)$ with $g' > 0$ on $(0, s^*)$ and $g' < 0$ on $(s^*, 1)$ (early self-amplification, late saturation). The canonical instance is the logistic $g(s) = r\,s(1 - s)$. That g depends on the accumulated state s is the operative content of self-reference (Axiom A0) on the damage coordinate; *(b)* is dissipation (Axiom A1). Finiteness $\mathbb{E}[\Sigma_\infty] \leq V(F_0)$ follows from $V \geq 0$ in L1.

Regularity Condition L3 (Asymmetry measure)**.** The stochastic-thermodynamic entropy production $\Delta\Sigma$ along any segment satisfies a detailed fluctuation relation $\mathbb{P}[\gamma^\dagger]/\mathbb{P}[\gamma] = e^{-\Delta\Sigma}$, where $\gamma^\dagger$ is the time-reversed segment of γ [9, 28].

Regularity Condition L4 (Return protocol and elasticity below yield)**.** *(L4a, return protocol.)* The return event $\mathcal{R}$ is realized as the reversed-crossing continuation of the crossing set $\mathcal{C}$ under the same monitoring protocol: $\mathcal{R} = \{\gamma^\dagger : \gamma \in \mathcal{C}\}$, with the reversal $\gamma \mapsto \gamma^\dagger$ measurable and involutive (distinct crossings reverse to disjoint events). *(L4b, elasticity below yield.)* Sub-threshold segments are *reverse-favorable* — below the yield point no irreversible deformation has occurred, so $\Delta\Sigma \leq 0$ on $\mathcal{C}^c$, strict on a set of positive probability; this suffices for $\mathbb{E}[(e^{-\Delta\Sigma} - 1)\mathbf{1}_{\mathcal{C}^c}] > 0$, the constitutive content of elasticity below the yield point.

2.4 The reduction theorem

The point of the axiomatization is that only A0–A1 are primitive: accumulation, transition, and irreversibility are *theorems* once the regularity conditions hold. Making the reduction explicit lets the load-bearing capacity of each axiom be tested on each substrate separately, and it is exactly this structure that produces the predicted null findings of a substrate that violates one regularity condition. These conditions are *constitutive postulates*, not merely technical hypotheses: in particular L2 is the load-bearing assumption from which the transition follows. The reduction is accordingly a *consistency* result—given the named postulates, accumulation, transition and irreversibility are not independent axioms—rather than a derivation from self-reference and dissipation alone.

Theorem 2.2 (Reduction)**.** *Under Axioms A0–A1:*

(i) (*Accumulation*) *L1 implies A2, and $\Sigma_t \uparrow \Sigma_\infty < \infty$ (via Axiom A1; L2 is reserved for (ii)).*

(ii) (*Transition, in expectation*) *L2 implies the mean form of A3: the expected cumulative profile $s(t) = \mathbb{E}[\Sigma_t]/\mathbb{E}[\Sigma_\infty]$ is sigmoid, with a unique inflection at the loading time $t_{\mathrm{y}} := s^{-1}(s^*)$ at which the damage rate peaks, $s(t_{\mathrm{y}}) = \hat{a} := s^*$ (the inflection* location $\hat{a}$, *distinct from the loading* time t_{y}*) — provided the onset satisfies* $0 < s(0) < s^*$*; if $s(0) \geq s^*$ the profile is purely concave (inflection at the origin: the "measured beyond yield" boundary case).*

(iii) (*Irreversibility*) *under the finite-window nondegeneracy* $0 < \mathbb{P}[\mathcal{C}] < 1$*, L3 with the return protocol (L4a) gives the unconditional reverse-event bound (needing* $\mathbb{P}[\mathcal{C}^c] > 0$*), and L3 with full L4 (L4a+L4b) implies the conditional form of A4 (needing* $\mathbb{P}[\mathcal{C}] > 0$*).*

Proof. **(i)** By Axiom A1 the conditional increments satisfy $\delta_t \geq 0$; by L1 they are nonnegative and adapted, so $\Sigma_{t+1} = \Sigma_t + \delta_t \geq \Sigma_t$ pathwise, which is Axiom A2; thus Σ_t is the predictable compensator of the nonnegative supermartingale $V(F_t)$. Taking expectations and telescoping, $\mathbb{E}[\Sigma_t] = \sum_{s<t} \mathbb{E}[\delta_s] = \mathbb{E}[V(F_0)] - \mathbb{E}[V(F_t)] \leq \mathbb{E}[V(F_0)] < \infty$, using $V \geq 0$ (L1). A nondecreasing process with uniformly bounded expectation converges almost surely by the monotone convergence theorem to a limit Σ_∞ with $\mathbb{E}[\Sigma_\infty] \leq \mathbb{E}[V(F_0)] < \infty$; hence $\Sigma_\infty < \infty$ a.s.

(ii) By L2 the mean damage $s(t) = \mathbb{E}[\Sigma_t]/\mathbb{E}[\Sigma_\infty]$ — the deterministic profile actually fitted — solves the autonomous equation $\dot{s} = g(s)$ from an onset $s(0) = s_0 \in (0,1)$. Since $g > 0$ on $(0,1)$, s is strictly increasing, and $g(1) = 0$ gives $s(t) \to 1$; the profile rises from s_0 to 1. Its curvature is $\ddot{s} = \frac{d}{dt} g(s) = g'(s)\, g(s)$, so $\operatorname{sign}(\ddot{s}) = \operatorname{sign}(g'(s))$ because $g > 0$ on the interior. By the single-peak property L2(c), g' changes sign exactly once, from $+$ to $-$, at s^*. If $s_0 < s^*$, strict monotonicity makes s reach the critical accumulated level $\hat{a} := s^*$ at exactly one loading time $t_{\mathrm{y}} := s^{-1}(s^*)$: the time-profile is convex before t_{y} and concave after, a sigmoid whose inflection sits at the accumulated level $\hat{a}$ (the normalized-damage *location*, distinct from the loading *time* t_{y} at which it is reached), where the expected damage rate peaks — the yield point (Axiom A3). If instead $s_0 \geq s^*$, the trajectory meets only $g' < 0$, so $\ddot{s} < 0$ throughout: the profile is purely concave with no interior inflection (a substrate measured beyond its yield onset). Crucially the inflection is *derived* from the state-dependence of g — read off the accumulated level s^* at which the rate peaks — rather than *posited* as a unimodality in time; this is what removes the circularity of inferring the inflection from an assumed time profile. The statement is at the level of the expected trajectory; the pathwise inflection of an individual realization, $\sigma(t) = \Sigma_t/\Sigma_\infty$ as in Axiom A3, is the empirically fitted quantity (one $\hat{a}$ per specimen) and agrees with the mean inflection under regularity of the increment process, but is not asserted here as a pathwise theorem.

(iii) Fix a finite observation window with onset below threshold ($s(0) < s^*$, as in (ii)), call γ a *crossing* if the normalized damage $\sigma = \Sigma/\Sigma_\infty$ exceeds $\hat{a}$ in the window (equivalently $\Sigma > \hat{a}\,\Sigma_\infty$), and write $\mathcal{C}$ for the set of crossings, with the named finite-window regularity $0 < \mathbb{P}[\mathcal{C}] < 1$ (nondegenerate noise $\sigma > 0$ with reachability is a sufficient condition for $\mathbb{P}[\mathcal{C}] > 0$; the finite window with onset below threshold leaves $\mathbb{P}[\mathcal{C}^c] > 0$, the diffusion staying sub-threshold with positive probability). By the detailed fluctuation relation L3, $\mathbb{P}[\gamma^\dagger] = e^{-\Delta\Sigma(\gamma)}\mathbb{P}[\gamma]$, so the integral fluctuation theorem $\mathbb{E}[e^{-\Delta\Sigma}] = 1$ holds [28] and the time-reverse of a crossing is realized with relative likelihood $e^{-\Delta\Sigma(\gamma)}$.

Here $\Delta\Sigma$ is the pathwise entropy production of L3, genuine rather than a reparametrization artifact in the history-carrying asymmetric case (L3 with the strict reverse-favorability of L4b), where $\gamma^\dagger \neq \gamma^{-1}$ (the history-carrying F_t, via A0, makes the reverse state path not the inverse of the forward operator; a reversible recursion would instead have $\gamma^\dagger = \gamma^{-1}$ and $\Delta\Sigma = 0$); it is *not* the mean damage rate g of L2, a distinct expectation-level object.

Unconditional form (from L3 and the return protocol L4a). By the return protocol L4(L4a), $\mathcal{R} = \{\gamma^\dagger : \gamma \in \mathcal{C}\}$ (the reversal involutive, so distinct crossings reverse to disjoint events), and L3 gives $\mathbb{P}[\mathcal{R}] = \sum_{\gamma\in\mathcal{C}} \mathbb{P}[\gamma^\dagger] = \mathbb{E}[e^{-\Delta\Sigma}\mathbf{1}_{\mathcal{C}}]$. Hence the probability that a crossing is undone is

$$\mathbb{P}[\mathcal{R}] = \mathbb{E}[e^{-\Delta\Sigma}\mathbf{1}_{\mathcal{C}}] = 1 - \mathbb{E}[e^{-\Delta\Sigma}\mathbf{1}_{\mathcal{C}^c}] < 1,$$

since $e^{-\Delta\Sigma} > 0$ and $\mathbb{P}[\mathcal{C}^c] > 0$. (The naive "$\Delta\Sigma > 0$ with positive probability $\Rightarrow \mathbb{E}[e^{-\Delta\Sigma}] < 1$" is invalid, since reversed paths can carry $\Delta\Sigma < 0$; the integral fluctuation theorem together with $\mathbb{P}[\mathcal{C}^c] > 0$ is what makes it rigorous.)

Conditional form (Axiom A4). By the return protocol L4(L4a), the return probability conditional on crossing — *defined* as the protocol-conditioned reverse likelihood over the crossing ensemble, $\mathbb{E}[e^{-\Delta\Sigma} \mid \mathcal{C}]$, rather than an ordinary forward conditional $\mathbb{P}[\mathcal{R}\cap\mathcal{C}]/\mathbb{P}[\mathcal{C}]$ — is

$$\mathbb{P}[\mathcal{R} \mid \mathcal{C}] = \mathbb{E}[e^{-\Delta\Sigma} \mid \mathcal{C}] = \frac{1 - \mathbb{E}[e^{-\Delta\Sigma}\mathbf{1}_{\mathcal{C}^c}]}{\mathbb{P}[\mathcal{C}]},$$

which is < 1 exactly when $\mathbb{E}[(e^{-\Delta\Sigma} - 1)\mathbf{1}_{\mathcal{C}^c}] > 0$, i.e. the sub-threshold segments are reverse-favorable. This is the formal counterpart of *elasticity below the yield point*: while $\Sigma \leq \hat{a}$ no irreversible yield has occurred, so the entropy production there is non-positive ($\Delta\Sigma \leq 0$ on $\mathcal{C}^c$, strict on a set of positive probability) and the net irreversibility is localized at and beyond the crossing. Under L4(L4b) (elasticity below yield) $\mathbb{P}[\mathcal{R} \mid \mathcal{C}] < 1$, which is Axiom A4; absent it, the unconditional bound from L3+L4a already gives $\mathbb{P}[\mathcal{R}] < 1$. ☐

Remark 2.3 (Physical grounding of L2). The single-peaked, state-dependent rate is the constitutive form of continuum damage mechanics, not an assumption tailored to the sigmoid: the damage variable accumulates irreversibly (Axiom A2) and its rate is read off the accumulated damage, not the clock. Normalizing $s = \Sigma/\Sigma_\infty$ to the terminal state makes $g(1) = 0$ structural — whether Σ_∞ is a fracture or a saturated non-equilibrium steady state, the normalized damage saturates at 1 — and the single-peaked profile is the generic damage signature (early self-amplification through defect interaction, late saturation as the load-bearing reserve is exhausted). Reading the rate off the accumulated state is what makes the transition (Theorem 2.2(ii)) follow rather than be posited: L2 carries an independent material referent, not a circular re-encoding of the inflection it implies.

Remark 2.4 (Where self-reference is load-bearing). Axiom A0 is not itself a step of the reduction; it is the physical source of the two constitutive properties that are. It supplies the state-dependence of the rate in L2 (the rate is driven by the self-generated accumulated state) and, in the history-carrying asymmetric case (L3/L4b), the non-involutivity $\gamma^\dagger \neq \gamma^{-1}$ used in irreversibility (Theorem 2.2(iii): the history-carrying state path is not the inverse of the forward operator). Outside these two inputs it carries no independent formal weight — the honest reading of its role, not a defect: self-reference enters the theory through L2 and the entropy-production asymmetry, and nowhere else.

Remark 2.5 (Nulls as violated regularity). Theorem 2.2 predicts its own failures. A substrate whose recovery balances damage ($\mathbb{E}[\delta_t] \to 0$, violating the strict-rise part of L2) shows no transition; a symmetric, reversible self-reference ($\gamma^\dagger = \gamma^{-1}$, $\Delta\Sigma = 0$, failing the strict asymmetry of L3/L4b) shows no irreversibility; an externally imposed sequence with no path asymmetry violates L3 as well. In each case the absence of a critical transition is a confirmed prediction of the reduction, not a counterexample to it.

3 Two-Scale Renormalization

A self-referential system is observed at several scales — token, turn, sequence, corpus, population — and the framework requires that the same structural law describe each. We make the inter-scale map precise.

Definition 3.1 (Renormalization map)**.** For scales $\ell < \ell'$ the renormalization map is the composition

$$M_{\ell\to\ell'} = N_{\ell'} \circ R_{\ell\to\ell'} \circ C_{\ell\to\ell'}, \tag{3}$$

where C *coarse-grains* micro-trajectories into blocks (windows, clusters, representative volumes), R *rescales* the effective parameters along the induced parameter flow $\theta_\ell \mapsto \theta_{\ell'}$, and N *normalizes* dimensionless invariants so that the renormalized description stands structurally on a par with the original.

Definition 3.2 (Normal form and scale-covariance ansatz)**.** We take the mean dynamics in the Onsager-type normal form

$$\dot{z} = F(z, \Sigma) - \nabla_z \Pi(z, \Sigma) + \Theta\, \nabla_z I(z), \tag{4}$$

with drift F, dissipation potential Π, and information coupling I: the gradient form of Assumption R1 in which $-\nabla_z\Pi$ is the dissipation of Axiom A1 and $\Theta\,\nabla_z I$ is the self-referential force of Axiom A0. We restrict attention to the *relevant sector*: the subspace of operators on which $M_{\ell\to\ell'}$ acts form-covariantly, returning (4) with renormalized $(F', \Pi', \Theta') = R_{\ell\to\ell'}(F, \Pi, \Theta)$.

Remark 3.3 (On the ansatz). Renormalization generically generates new operators, so exact form-closure is not automatic; the restriction to the relevant sector is the standard content of a renormalization-group analysis, in which irrelevant operators are suppressed under coarse-graining and the surviving form-covariant subspace carries the universal structure [15, 32]. Within this sector the construction is consistent: coarse-graining C replaces z by a block average whose fluctuations contribute, to leading order, a renormalized diffusion absorbed into Π' (Assumption R4), and each gradient term pulls back to a gradient term under the smooth change of variables induced by R. Downstream results use only this covariance, not a claim that every microscopic dynamics lies in the relevant sector.

The five operative scales are: tokens (ℓ_0), turn-pairs (ℓ_1), the sequence (ℓ_2, carrier of the inflection $\hat{a}$ and the accumulator), the corpus (ℓ_3, carrier of the $\hat{a}$-distribution), and the population (ℓ_4). The scale-covariance of Definition 3.2 is what licenses transporting the order parameter $\hat{a}$ across them.

4 The Operational Frame

The abstract class of Section 2 is instantiated, at the sequence scale ℓ_2, by a measurement system on embedded trajectories. The instantiation is metric-agnostic in principle; we record the current operational definitions so that the abstract objects $(\Sigma, \hat{a})$ have explicit observable counterparts.

Definition 4.1 (Deformation observables)**.** Let $(z_t)_{t\leq T}$, $z_t = (u_t, a_t) \in \mathbb{R}^{2d}$, be the embedded turn-pair sequence and $v_t := \|z_t - z_{t-1}\|$ the semantic velocity. Five coupled deformation axes are defined on windows $\mathcal{W}(t)$:

(D1) **Stiffness** $E_a := 1/\operatorname{Var}_{\mathcal{W}}(v)$, inverse velocity variance (strain hardening);

(D2) **Asymmetry** MAS, the meta-asymmetry between the user and assistant sub-trajectories in $\mathbb{R}^d$ (one-sided load introduction, cantilever);

(D3) **Accumulation** B_w, the windowed budget accumulator, a monotone functional of the cumulative geometric displacement (creep/fatigue);

(D4) **Brittleness** dir_consistency, the directional coherence of successive increments, whose collapse marks brittle break-out;

(D5) **Coupling** $\operatorname{xcorr}(v, w)$, the cross-correlation of semantic velocity v and a structural displacement field w (dynamics–structure covariation).

The order parameter is the **inflection** $\hat{a}$: the location of the unique inflection (Theorem 2.2(ii)) of the sigmoid fit of cumulative damage Σ against cumulative strain.

Remark 4.2 (Anti-sparse coupling)*.* The five axes are coupled, not redundant. In the linear-response (Hooke) regime $\sigma = E\varepsilon$ ties stress, stiffness and strain analytically; the same coupling recurs among the axes, so stiffness (D1) and accumulation (D3) are two projections of one material response. A sparsity-inducing (ℓ_1) selection would discard such coupled coordinates as collinear and thereby lose information, not noise: the correct paradigm is materials testing, where coupled gauges record one response from distinct axes, not feature selection over independent predictors. Consistently, the loading-time autocorrelation of the cumulative family is signal: measurements along a deformation curve are not independent, and specimen-level inference must use the whole curve as the unit (block resampling), not the individual measurement.

Definition 4.3 (Budget state machine)**.** The accumulator B_w drives a monotone hysteretic state machine with ordered regimes Normal $\preceq$ Watch $\preceq$ Warning $\preceq$ Critical, transitions of which are governed by policy thresholds and a recovery rate. By the No-Go theorem (Section 9) these thresholds are calibration parameters, not universal constants; only the *architecture* (monotone accumulation with hysteretic, asymmetric transitions, Axioms A2,A4) is part of the theory.

The embedding instrument is a sentence encoder satisfying Assumption R3 [25]; decreasing embedding variance signals stiffening of the semantic space, and irreversibly growing cumulative displacement documents plastic deformation, the observable face of Axioms A2 and A4.

5 T1: Information–Stability Correspondence

The first structural theorem links dynamical stability, measured by the conditional phase-space volume rate (the sum of conditional Lyapunov exponents), to the trend of mutual information between the coupled coordinates $p := u$ (prompt/user) and $r := a$ (response/assistant), $z = (p, r)$.

Lemma 5.1 (Conditional log-volume rate)**.** *Work over a finite transient window on which the local linearization $A := DF|_{r|p}$ is constant (Assumptions R1–R3) and the conditional covariance $\Sigma_{r|p}(t) \succ 0$ obeys the conditional Lyapunov equation $\dot{\Sigma}_{r|p} = A\,\Sigma_{r|p} + \Sigma_{r|p}A^\top + \sigma^2 I$, with the leading*

Oseledets mode transverse to the conditioning block p *(it has a component in the conditional response subspace, not annihilated by conditioning on p). The conditional log-volume then obeys the exact rate*

$$\frac{d}{dt}\,\tfrac{1}{2}\log\det\Sigma_{r|p} = \mathrm{Tr}(A) + \tfrac{1}{2}\sigma^2\,\mathrm{Tr}\,(\Sigma_{r|p}^{-1}),$$

whose first term is the conditional phase-space volume rate $\mathrm{Tr}(A) = \sum_i \lambda_i$ *(the sum of the conditional Lyapunov exponents, i.e. minus the conditional dissipation rate) and whose second is the noise-injection rate.* (Expanding case.) *If* $\mathrm{Tr}(A) > 0$ *both terms are positive, so the log-volume strictly grows.* (Dissipative case.) *If A is Hurwitz (*$\mathrm{Tr}(A) < 0$*, with stationary* $\Sigma_\infty \succ 0$ *solving* $A\Sigma_\infty + \Sigma_\infty A^\top + \sigma^2 I = 0$*), then from an excited start* $\Sigma_{r|p}(0) \succ \Sigma_\infty$ *the log-volume decreases monotonically toward* $\frac{1}{2}\log\det\Sigma_\infty$. (General transient.) *More generally, over a window on which the noise-injection term is subdominant (large* $\Sigma_{r|p}$*), the windowed log-volume slope carries the sign of* $\mathrm{Tr}(A)$ *— including dissipative chaos, where one expanding direction (*$\lambda_{\max} > 0$*) coexists with net volume contraction (*$\mathrm{Tr}(A) < 0$*).* (Dominant-mode case.) *When the sub-leading exponents are sign-negligible,* $\left|\sum_{i>1}\lambda_i\right| < |\lambda_{\max}|$ *with* $\lambda_{\max} \neq 0$ *(e.g. a one-dimensional conditional response subspace), then* $\operatorname{sgn}\mathrm{Tr}(A) = \operatorname{sgn}\lambda_{\max}$*, recovering the classical maximal-exponent reading. (At stationarity the rate vanishes; these are transient statements.)*

Proof. Rate. For $\Sigma_{r|p} \succ 0$, Jacobi's formula gives $\frac{d}{dt}\frac{1}{2}\log\det\Sigma_{r|p} = \frac{1}{2}\mathrm{Tr}(\Sigma_{r|p}^{-1}\dot{\Sigma}_{r|p})$; substituting the Lyapunov equation and using $\mathrm{Tr}(\Sigma_{r|p}^{-1}A\Sigma_{r|p}) = \mathrm{Tr}(A) = \mathrm{Tr}(A^\top)$ yields $\mathrm{Tr}(A) + \frac{1}{2}\sigma^2\,\mathrm{Tr}(\Sigma_{r|p}^{-1})$. By Liouville's identity $\mathrm{Tr}(A) = \sum_i \lambda_i$ is the divergence of the conditional flow. The expanding case is immediate (both terms positive). *Dissipative case.* Pass to the stationary gauge $G := \Sigma_\infty^{-1/2}(\Sigma_{r|p} - \Sigma_\infty)\Sigma_\infty^{-1/2} \succeq 0$ and $\tilde{A} := \Sigma_\infty^{-1/2}A\Sigma_\infty^{1/2}$; the stationary Lyapunov equation becomes $\tilde{A} + \tilde{A}^\top = -\sigma^2\Sigma_\infty^{-1} \prec 0$, and $\dot{E} = AE + EA^\top$ for $E := \Sigma_{r|p} - \Sigma_\infty$ transports to $\dot{G} = \tilde{A}G + G\tilde{A}^\top$. Then, since G and $(I+G)^{-1}$ commute,

$$\frac{d}{dt}\,\tfrac{1}{2}\log\det\Sigma_{r|p} = \tfrac{1}{2}\mathrm{Tr}\big((I+G)^{-1}\dot{G}\big) = \tfrac{1}{2}\mathrm{Tr}\big((I+G)^{-1}G(\tilde{A}+\tilde{A}^\top)\big) = -\tfrac{1}{2}\sigma^2\,\mathrm{Tr}\big((I+G)^{-1}G\,\Sigma_\infty^{-1}\big) \leq 0,$$

the trace of a product of the positive-semidefinite $(I+G)^{-1}G$ and the positive-definite Σ_∞^{-1} being nonnegative, strict unless $G = 0$; with the excited start $G(0) \succ 0$ the log-volume decreases monotonically to $\frac{1}{2}\log\det\Sigma_\infty$. *General transient.* Where $\frac{1}{2}\sigma^2\,\mathrm{Tr}(\Sigma_{r|p}^{-1})$ is subdominant (large $\Sigma_{r|p}$), the rate, and hence the windowed slope, takes the sign of $\mathrm{Tr}(A)$. *Dominant mode.* If $\left|\sum_{i>1}\lambda_i\right| < |\lambda_{\max}|$ with $\lambda_{\max} \neq 0$ (e.g. a one-dimensional conditional response subspace), then $\operatorname{sgn}\sum_i\lambda_i = \operatorname{sgn}\lambda_{\max}$. Throughout we use the *conditional* covariance $\Sigma_{r|p}$, not the single-time state covariance: it is the perturbation covariance the Lyapunov spectrum governs. □

Lemma 5.2 (Two-sided entropy bounds)**.** *Let* $X \in \mathbb{R}^n$ *have covariance* Σ_X*. Then* $h(X) \leq \frac{1}{2}\log\big((2\pi e)^n\det\Sigma_X\big)$*, with equality iff X is Gaussian [8, Thm. 8.6.5]. If moreover the law of X is log-concave (Assumption R4), there is a dimensional constant* $c_n \in \mathbb{R}$ *with* $h(X) \geq \frac{1}{2}\log\det\Sigma_X + c_n$*.*

Proof. The upper bound is the Gaussian maximum-entropy property. For the lower bound, reduce to the isotropic case by $Y := \Sigma_X^{-1/2}(X - \mathbb{E}X)$, which is log-concave with identity covariance and satisfies $h(X) = h(Y) + \frac{1}{2}\log\det\Sigma_X$. Isotropic log-concave laws obey a uniform entropy lower bound $h(Y) \geq c_n$ by the variance–entropy comparison from the Brascamp–Lieb inequality and the concentration of log-concave measures [6, 5]. Combining gives the claim. □

Theorem 5.3 (T1: Information–stability correspondence via the conditional volume rate (finite-window slope))**.** *Assume R1, R2, R3, R5 and the concentration consequence of R4, but* not *the full joint stationarity of (p_t, r_t) on the window (which would render I time-invariant). Assume a* local-linear, exactly conditional-Gaussian, homoscedastic *model over a finite transient window W on which $A := DF|_{r|p}$ is constant: $\pi(r \mid p)$ is Gaussian with covariance $\Sigma_{r|p} \succ 0$* independent of the value of p*, obeying the conditional Lyapunov equation, and the marginal $h(r)$ stationary (bounded). Write $V_c := \frac{1}{2}\log\det\Sigma_{r|p}$ and let $\operatorname{slope}_W f$ be the least-squares slope of f over W. Assume two named hypotheses:*

(D) slope-dominance (identifiability): $\left|\operatorname{slope}_W h(r)\right| < \left|\operatorname{slope}_W V_c\right|$*;*

($\mathrm{H_{vol}}$) volume-rate regime: *the window lies in a sign-valid regime of Lemma 5.1, i.e.* $\operatorname{sgn}(\operatorname{slope}_W V_c) = \operatorname{sgn}(\operatorname{Tr} A)$*; it suffices that* $\operatorname{Tr}(A) > 0$*; or that A is Hurwitz with $\Sigma_{r|p}(t) \succeq \Sigma_\infty$ throughout W and $\Sigma_{r|p} \not\equiv \Sigma_\infty$ on W (strictly excited start); or that the noise-injection term is quantitatively subdominant, $\frac{1}{2}\sigma^2 \operatorname{Tr}(\Sigma_{r|p}^{-1}) < |\operatorname{Tr} A|$ on W (when* $\operatorname{Tr} A < 0$*).*

Then over W,

$$\operatorname{sgn}\big(\operatorname{slope}_W I\big) = -\operatorname{sgn}\big(\operatorname{slope}_W V_c\big) \overset{(\mathrm{H_{vol}})}{=} -\operatorname{sgn}\big(\operatorname{Tr}(A)\big), \qquad I(t) := I(p_t; r_t),$$

the first equality exact, the second under (H_{vol}); no sign is claimed for $\operatorname{Tr}(A) = 0$ *or for covariance-below-stationary transients. Equivalently, the coupling rises exactly when the conditional uncertainty volume contracts (conditional dissipation) and falls when it expands.*

Proof. By the exact homoscedastic conditional-Gaussian law, $h(r \mid p) = \frac{1}{2}\log\det(2\pi e\,\Sigma_{r|p}) = V_c + \frac{n}{2}\log(2\pi e)$ identically: the additive constant has *zero* windowed slope. (This exact-Gaussian assumption is what replaces the merely two-sided bound of Lemma 5.2: the entropy deficit is constant, not just bounded, so it cannot carry an affine windowed trend.) Hence with $I = h(r) - h(r \mid p)$ and the windowed slope linear, $\operatorname{slope}_W I = \operatorname{slope}_W h(r) - \operatorname{slope}_W V_c$. By slope-dominance (D), $|\operatorname{slope}_W h(r)| < |\operatorname{slope}_W V_c|$, so $\operatorname{sgn}(\operatorname{slope}_W I) = -\operatorname{sgn}(\operatorname{slope}_W V_c)$, the exact first equality. By ($\mathrm{H_{vol}}$) and Lemma 5.1, $\operatorname{sgn}(\operatorname{slope}_W V_c) = \operatorname{sgn}(\operatorname{Tr} A)$ in a sign-valid regime (exactly, with monotone decrease, in the Hurwitz/excited case via the stationary-gauge computation), giving the second equality. At $\operatorname{Tr}(A) = 0$ the positive noise term leaves the volume slope without a definite sign, and at stationarity $\Sigma_{r|p}$ and hence I are time-invariant; neither is claimed. The argument uses the conditional entropy and conditional covariance directly; it does not infer mutual information from a cross-covariance coefficient. □

Corollary 5.4 (Maximal-exponent reading)**.** *Under the dominant-mode hypothesis of Lemma 5.1 ($|\sum_{i>1}\lambda_i| < |\lambda_{\max}|$, $\lambda_{\max} \neq 0$, so* $\operatorname{sgn}\operatorname{Tr}(A) = \operatorname{sgn}\lambda_{\max}$*) together with ($H_{\mathrm{vol}}$), the correspondence becomes* $\operatorname{sgn}(\lambda_{\max}) = -\operatorname{sgn}(\operatorname{slope}_W I)$*: the classical maximal-Lyapunov reading is the dominant-mode special case.*

Remark 5.5 (Conditional, not cross-covariance). The correspondence is read off the *conditional* entropy $h(r \mid p)$ through its exact Gaussian value (Lemma 5.2 bounding it in the general log-concave case) and the conditional log-volume rate of Lemma 5.1. It deliberately avoids inferring mutual information from a cross-covariance coefficient (valid only in the Gaussian case) and from the single-time state covariance; the log-concave conditional law is exactly the structure that makes the entropy bounds two-sided. The earlier non-degeneracy condition (N) is thereby subsumed into the explicit conditional hypotheses.

Remark 5.6 (Finite-window diagnostic; volume rate, not maximal exponent)*.* Two scope conditions make T1 a finite-time *diagnostic* rather than an asymptotic law. First, the stability signature is the *conditional volume rate* $\mathrm{Tr}(A) = \sum_i \lambda_i$ (the conditional Lyapunov-exponent sum, i.e. minus the conditional dissipation rate), not the maximal exponent $\lambda_{\max}$: the log-volume tracks net phase-space contraction, so a system with one expanding direction but net contraction ($\lambda_{\max} > 0$, $\mathrm{Tr}(A) < 0$ — dissipative chaos collapsing onto an attractor) correctly shows *rising* coupling. The maximal-exponent reading (Corollary 5.4) is the dominant-mode special case. The empirical coupling is the finite-window average $\mathrm{VE}(t) = \frac{1}{W}\sum_{i\in\mathcal{W}(t)} I(p_i; r_i)$, and the correspondence is between its windowed slope and the transient volume rate, carrying a Cesàro/finite-window approximation that must be quantified per application. Second, the conditional Lyapunov equation of Lemma 5.1 is the *local-linearization* (Assumption R2 with the bi-Lipschitz embedding R3) of the perturbation covariance; it governs the transient, not the stationary law, where $\Sigma_{r|p}$ and I are time-invariant. Within this scope a declining windowed coupling signals conditional volume *expansion* (transient, local instability) and a rising coupling signals *contraction* (dissipation toward an attractor) — the abstract content of axis (D5) of Definition 4.1, and the observable face of the dissipation axiom A1; it is not asserted as an identity at stationarity. The slope-dominance hypothesis of Theorem 5.3 is a measurement (identifiability) condition — the conditional log-volume signal must exceed the marginal-entropy variation over the window — not a covert encoding of the conclusion's sign, in the same spirit as the non-degeneracy hypothesis $q_0''(1/2) < 0$ of Theorem 9.4.

6 T2: Autopoietic Stationarity

The second theorem gives conditions under which the operator recursion of Definition 2.1 has a unique, globally attracting non-equilibrium steady state. It requires one structural hypothesis beyond the axioms.

Definition 6.1 (Contraction condition)**.** The meta-operator Φ acts on a complete metric space $(\mathcal{F}, d_{\mathcal{F}})$ as a κ-contraction in its first argument: there is $\kappa \in (0,1)$ with $d_{\mathcal{F}}(\Phi(F_1,\cdot),\Phi(F_2,\cdot)) \le \kappa\, d_{\mathcal{F}}(F_1,F_2)$ for all F_1, F_2. We take the induced *autonomous* self-map $\bar{\Phi}(F) := \Phi(F, z^*, u^*)$ explicitly (frozen at the stationary (z^*, u^*), or averaged over a stationary input law, or a specified closed-loop selector), to which the fixed-point argument applies, with the autopoiesis-compatibility $\bar{\Phi}(F^*) = F^*$. Moreover the stationary state drift is *two-sided dissipative* about its drift equilibrium z^* (the point $F^*(z^*) = 0$): there exist $0 < c \le c' < \infty$ with $-c'\,\|z - z^*\|^2 \le \langle z - z^*, F^*(z)\rangle \le -c\,\|z - z^*\|^2$ globally on $\mathbb{R}^m$ (or an explicitly invariant domain), with the noise non-degenerate and uniformly elliptic and the dynamics non-explosive. (These are hypotheses on the *state* dynamics; Axiom A1 concerns the operator-level functional and does not by itself bound F^*.)

Theorem 6.2 (T2: Autopoietic stationarity)**.** *Under Axioms A0–A1 and Definition 6.1:*

(i) *There is a unique autopoietic fixed point F^* with $\bar{\Phi}(F^*) = F^*$ (equivalently $\Phi(F^*, z^*, u^*) = F^*$), and the iterates $F_{n+1} = \bar{\Phi}(F_n)$ converge geometrically, $d_{\mathcal{F}}(F_n, F^*) \le \frac{\kappa^n}{1-\kappa}\, d_{\mathcal{F}}(F_1, F_0)$.*

(ii) *The state diffusion under F^* is geometrically ergodic, with a unique, globally attracting invariant law μ — a non-equilibrium steady state when the stationary probability current $J_\mu = F^*\mu - \frac{\sigma^2}{2}\nabla\mu$ does not vanish (detailed balance fails), otherwise a non-degenerate stationary law; its stationary fluctuation level $\bar{V} = \mathbb{E}_\mu[\frac{1}{2}\,\|z - z^*\|^2]$ is bounded away from zero, in the band $[\,\sigma^2 m/(4c'),\, \sigma^2 m/(4c)\,]$.*

(iii) *At F^*, for an integrable z-observable (or jointly stationary (z, u); Definition 6.1), the dissipation functional is stationary, $\dot{D} = 0$.*

Proof. **(i)** $(\mathcal{F}, d_{\mathcal{F}})$ is complete and Φ is a κ-contraction (Definition 6.1); the Banach fixed-point theorem [2] yields a unique F^* with $\Phi(F^*) = F^*$ and, for any F_0, the a-priori estimate $d_{\mathcal{F}}(F_n, F^*) \le \frac{\kappa^n}{1-\kappa} d_{\mathcal{F}}(F_1, F_0)$.

(ii) Fix the operator at F^*; the state then solves the autonomous diffusion $dz = F^*(z)\,dt + \sigma\,dW_t$ (Assumption R1), with z^* the drift equilibrium $F^*(z^*) = 0$ of Definition 6.1. The two-sided dissipativity is a Foster–Lyapunov drift condition and the noise is non-degenerate ($\sigma > 0$); the diffusion is therefore non-explosive and geometrically ergodic, with a unique invariant law μ to which the law of $z(t)$ converges—the unique, globally attracting non-equilibrium steady state [17, 19]. To quantify its fluctuation level, define the quadratic Lyapunov functional $\mathcal{V}(z) := \frac{1}{2}\|z - z^*\|^2$. By Itô's formula,

$$d\mathcal{V} = \langle z - z^*,\, F^*(z)\rangle\,dt + \tfrac{1}{2}\sigma^2 m\,dt + \sigma\langle z - z^*, dW_t\rangle.$$

The two-sided dissipativity of Definition 6.1 gives $-2c'\,\mathcal{V}(z) \le \langle z - z^*, F^*(z)\rangle \le -2c\,\mathcal{V}(z)$. Write $\bar{V}(t) := \mathbb{E}[\mathcal{V}(z(t))]$; taking expectations and using the Itô noise injection rate $\frac{1}{2}\sigma^2 m$, where m is the state dimension of Assumption R1, the two bounds sandwich the mean Lyapunov rate:

$$-2c'\,\bar{V} + \tfrac{1}{2}\sigma^2 m \;\le\; \dot{\bar{V}} \;\le\; -2c\,\bar{V} + \tfrac{1}{2}\sigma^2 m.$$

Both scalar comparison ODEs are globally attracting (to $\sigma^2 m/(4c')$ and $\sigma^2 m/(4c)$ respectively), so by comparison

$$\liminf_{t\to\infty} \bar{V}(t) \ge \tfrac{\sigma^2 m}{4c'} > 0, \qquad \limsup_{t\to\infty} \bar{V}(t) \le \tfrac{\sigma^2 m}{4c}.$$

The band $[\sigma^2 m/(4c'), \sigma^2 m/(4c)]$ is asymptotically absorbing (the limit exists when $c = c'$). The $\liminf$ bound keeps the stationary fluctuation level bounded away from 0 (the system maintains finite fluctuations rather than collapsing to a deterministic point, non-equilibrium); absorption into the band is global attraction, stability.

(iii) Let $D := \mathbb{E}[C(z)] - C_{\mathrm{eq}}$ be the dissipation functional, with $C = C(z)$ a z-observable, integrable ($C \in L^1(\mu)$, no explicit time dependence) whose minimization is the gradient structure of (4) (if C depends jointly on (z, u), assume (z_t, u_t) jointly stationary under F^* — e.g. $u \equiv u^*$ fixed or an independent stationary input). Then $\dot{D} = \partial_t \mathbb{E}[C]$. In the stationary regime of (ii) the law of z is time-invariant, so $\partial_t \mathbb{E}[C] = 0$ for such an observable. Hence $\dot{D} = 0$: entropy production is exactly balanced by dissipative export, the defining balance of a non-equilibrium steady state [24]. □

Remark 6.3. T2 is the existence statement behind the autopoietic configuration of Definition 2.1: self-reference plus dissipation plus contraction produces a stable far-from-equilibrium attractor, not a dead equilibrium. The geometric rate in (i) is conditional on a *uniform* modulus $\kappa < 1$; if the per-step contraction degrades ($\kappa_n \to 1$), convergence is governed by the cumulative budget rather than a fixed rate (Section 12).

7 T3: The Goldilocks Principle

The third result concerns where, relative to criticality, a self-referential system functions best. It is stated as a hypothesis, with a proved partial result; the hypothesis status is intrinsic, because the performance functional is not fixed by the axioms.

Hypothesis 7.1 (Goldilocks principle)**.** Let $L(\mathrm{I}, \mathrm{C})$ be a performance functional in innovation I (exploration/variety) and coherence C (stability/recurrence), jointly concave with positive cross-derivative $\partial^2 L/\partial \mathrm{I}\,\partial \mathrm{C} > 0$. Then optimal performance requires interior balance, neither $\mathrm{I} = 0$

nor $\mathrm{C} = 0$, and the optimum lies in a neighbourhood of criticality $|\lambda_{\max}| \approx 0$ rather than at the measure-zero point of exact criticality.

Proposition 7.2 (Necessity of interior balance)**.** *Let $L : [0,\infty)^2 \to \mathbb{R}$ be C^2 and strictly concave, with positive cross-partial $\partial^2 L/\partial \mathrm{I}\, \partial \mathrm{C} > 0$ and strictly positive boundary gradients $\partial_{\mathrm{I}} L(0,0) > 0$, $\partial_{\mathrm{C}} L(0,0) > 0$. If L attains a maximum on $[0,\infty)^2$, it is at a unique interior point $(\mathrm{I}^*, \mathrm{C}^*)$ with $\mathrm{I}^* > 0$, $\mathrm{C}^* > 0$, and every boundary point is strictly suboptimal.*

Proof. A strictly concave function on a convex domain has at most one local maximum, which is then global. At an interior maximum the first-order conditions $\partial_{\mathrm{I}} L = \partial_{\mathrm{C}} L = 0$ hold. Suppose the maximum were on the boundary, say at $\mathrm{I} = 0$; optimality on that face requires $\partial_{\mathrm{I}} L \leq 0$ along $\mathrm{I} = 0$. But by the positive cross-partial and the strict boundary-gradient hypotheses, for every $\mathrm{C} \geq 0$,

$$\partial_{\mathrm{I}} L(0, \mathrm{C}) = \partial_{\mathrm{I}} L(0,0) + \int_0^{\mathrm{C}} \partial^2_{\mathrm{IC}} L \, d\mathrm{C}' \;\geq\; \partial_{\mathrm{I}} L(0,0) \;>\; 0,$$

since the integrand is positive (the integral is ≥ 0, vanishing only at the corner $\mathrm{C} = 0$). This contradicts boundary optimality on the entire face $\mathrm{I} = 0$, the corner $(0,0)$ included; the maximum is therefore interior with $\mathrm{I}^* > 0$, and symmetrically $\mathrm{C}^* > 0$. Strict concavity makes the interior critical point the unique global maximum, with every boundary point strictly below it. ☐

Remark 7.3 (Why a hypothesis). Proposition 7.2 proves the interior-balance *structure*; the full Hypothesis 7.1 remains open because (a) the functional L is application-dependent and not derivable from the axioms, and (b) the identification of the interior optimum with the neighbourhood of $\lambda_{\max} = 0$ is an empirical prediction: systems operating near, but not at, criticality outperform both rigid (C saturated, $\lambda_{\max} \ll 0$) and chaotic ($\lambda_{\max} \gg 0$) regimes. The principle is the optimization counterpart of T1: criticality (the edge $\lambda_{\max} \approx 0$, the dominant-mode reading of the conditional volume rate) is the boundary where the information–stability sign flips, and function concentrates in its neighbourhood.

8 The Renormalization Operator and the Inflection Fixed Point

We now realize the inflection $\hat{a}$ of Axiom A3 as the order parameter of a concrete operator fixed point, and quantify its convergence. We make the construction self-contained.

8.1 The cone, the smoothing channel, and strict positivity

Let

$$\mathcal{K} := \{f \in C([0,1]) : f \geq 0,\ f \text{ non-increasing}\}, \qquad \mathcal{M} := \{Q \in \mathcal{K} : Q(0) = 1,\ Q(1) = 0\}$$

be the closed convex cone of non-negative non-increasing profiles and the anchored *survival* class ($\mathcal{M}$ consists of the survival functions $1 - F$ of probability measures on $[0,1]$; the right anchor $Q(1) = 0$ is intrinsic, not optional). The damped operator below (Definition 8.3) acts on a gap-bounded admissible subclass $\mathcal{U}_\rho \subset \mathcal{M}$. For strictly positive $f, g \gg 0$ the Hilbert projective metric is

$$d_H(f,g) := \ln \sup_{x\in[0,1]} \frac{f(x)}{g(x)} - \ln \inf_{x\in[0,1]} \frac{f(x)}{g(x)},$$

which is scale-invariant: $d_H(\alpha f, \beta g) = d_H(f,g)$ for $\alpha, \beta > 0$. A kernel $k : [0,1]^2 \to (0,\infty)$ is *totally positive of order two* (TP2) if $k(x_1,y_1)k(x_2,y_2) \geq k(x_1,y_2)k(x_2,y_1)$ for $x_1 < x_2$, $y_1 < y_2$;

the normalized smoothing operator is $(T_\varepsilon f)(x) := \int_0^1 \tilde{k}_\varepsilon(x,y)f(y)\,dy$ with $\tilde{k}_\varepsilon(x,y) = k_\varepsilon(x,y)/Z_\varepsilon(x)$, $Z_\varepsilon(x) = \int_0^1 k_\varepsilon(x,y')\,dy'$, the prototype being the Gaussian $k_\varepsilon(x,y) = e^{-(x-y)^2/\varepsilon}$.

Lemma 8.1 (Instant regularization)**.** *For every $\varepsilon > 0$ and every $f \in \mathcal{K} \setminus \{0\}$, the smoothed profile $T_\varepsilon f$ is strictly positive on all of $[0,1]$ (endpoints included) and non-increasing. Consequently d_H is well defined on every image $T_\varepsilon f$, even when f vanishes at an endpoint.*

Proof. Strict positivity: since $k_\varepsilon(x,y) > 0$ for all x, y and $f \geq 0$ with $f \not\equiv 0$, $(T_\varepsilon f)(x) = \int_0^1 \tilde{k}_\varepsilon(x,y)f(y)\,dy > 0$ for every $x \in [0,1]$, the endpoints included. Monotonicity is the variation-diminishing property of TP2 kernels [16]: a TP2 kernel maps non-increasing functions to non-increasing functions, and row normalization (division by $Z_\varepsilon(x)$, which depends on x only) preserves the TP2 cross-ratio. The endpoint values, where an anchored profile of CDF type may vanish, are thereby lifted strictly above 0, so the Hilbert metric—defined only for $f \gg 0$—applies to the entire image of the channel. $\square$

Proposition 8.2 (Explicit finite projective diameter of the Gaussian channel)**.** *For $k_\varepsilon(x,y) = e^{-(x-y)^2/\varepsilon}$ the projective diameter of the image cone under T_ε, over the full positive cone, is finite and computed exactly:*

$$\Delta(T_\varepsilon) = \sup_{x,y,u,v\in[0,1]} \ln \frac{k_\varepsilon(x,y)\,k_\varepsilon(u,v)}{k_\varepsilon(x,v)\,k_\varepsilon(u,y)} = \frac{2}{\varepsilon}.$$

Proof. The Gaussian form gives

$$\ln \frac{k_\varepsilon(x,y)k_\varepsilon(u,v)}{k_\varepsilon(x,v)k_\varepsilon(u,y)} = -\tfrac{1}{\varepsilon}\left[(x-y)^2 + (u-v)^2 - (x-v)^2 - (u-y)^2\right] = \tfrac{2}{\varepsilon}(x-u)(y-v),$$

which on $[0,1]^4$ attains its maximum at the corner $(x,u,y,v) = (1,0,1,0)$, giving $\Delta(T_\varepsilon) = 2/\varepsilon < \infty$. This supremum is taken over the full positive cone; restricted to the monotone cone of nonincreasing profiles the diameter is no larger, so the Birkhoff coefficient $\kappa(\varepsilon) \leq \tanh(1/(2\varepsilon))$ used below is an upper bound on the contraction rate (sufficient for convergence, possibly not attained on the monotone subclass). $\square$

Definition 8.3 (Anchoring gauge, admissible class, damped operator)**.** With the Gaussian smoothing channel T_ε as above, write $\tilde{N}(Q) := (T_\varepsilon Q)(0) - (T_\varepsilon Q)(1)$ for the post-smoothing gap, fix a non-degeneracy threshold $\rho > 0$, and let $\mathcal{U}_\rho := \{Q \in \mathcal{M} : \tilde{N}(Q) \geq \rho\}$ be the admissible class. The *anchoring gauge* — a scale-free two-end projective normalization, the analogue of $x \mapsto x/\|x\|$ — is

$$\mathcal{A}(Q) := \frac{T_\varepsilon Q - (T_\varepsilon Q)(1)}{\tilde{N}(Q)}, \qquad \mathcal{A}(Q)(0) = 1,\ \mathcal{A}(Q)(1) = 0,$$

and the *damped renormalization operator*, damping toward the reference profile $Q_{\mathrm{ref}}(x) := 1 - x$ with weight $\eta \in (0,1)$, is

$$\mathcal{R}_{\eta,\varepsilon}(Q) := (1-\eta)\,\mathcal{A}(Q) + \eta\, Q_{\mathrm{ref}}.$$

Lemma 8.4 (Anchoring self-map)**.** *For $Q \in \mathcal{U}_\rho$ the anchored profile $\mathcal{A}(Q)$ is nonincreasing with $\mathcal{A}(Q)(0) = 1$ and $\mathcal{A}(Q)(1) = 0$. Since $Q_{\mathrm{ref}} \in \mathcal{U}_\rho$, the operator $\mathcal{R}_{\eta,\varepsilon}$ maps $\mathcal{U}_\rho$ into itself.*

Proof. By Lemma 8.1 $T_\varepsilon Q$ is nonincreasing, hence so is $\mathcal{A}(Q)$; the endpoint values follow from the definition of $\mathcal{A}$. The gap $\tilde{N}(\mathcal{A}(Q)) = \int_0^1 (\tilde{k}_\varepsilon(0,y) - \tilde{k}_\varepsilon(1,y))\mathcal{A}(Q)(y)\,dy \geq 0$ because $\mathcal{A}(Q)$ is nonincreasing and T_ε preserves monotonicity (no lower bound on this gap is needed: the admissibility gap of $\mathcal{R}_{\eta,\varepsilon}(Q)$ is supplied by the reference profile below). As $\mathcal{A}(Q)$ and $Q_{\mathrm{ref}}(x) = 1 - x$ are nonincreasing with the two-end anchor, $\mathcal{R}_{\eta,\varepsilon}(Q) = (1-\eta)\mathcal{A}(Q) + \eta Q_{\mathrm{ref}}$ is again nonincreasing and two-end-anchored. Its admissibility gap is linear in the two parts (the gap functional $\tilde{N}$ is linear, T_ε being linear):

$$\tilde{N}(\mathcal{R}_{\eta,\varepsilon}(Q)) = (1-\eta)\,\tilde{N}(\mathcal{A}(Q)) + \eta\,\tilde{N}(Q_{\mathrm{ref}}) \;\geq\; \eta\,\tilde{N}(Q_{\mathrm{ref}}) \;>\; 0,$$

where $\tilde{N}(Q_{\mathrm{ref}}) = (T_\varepsilon Q_{\mathrm{ref}})(0) - (T_\varepsilon Q_{\mathrm{ref}})(1) > 0$ because the strictly positive Gaussian channel maps the strictly decreasing $1-x$ to a strictly decreasing profile. Choosing $\rho \leq \eta\,\tilde{N}(Q_{\mathrm{ref}})$ gives $\mathcal{R}_{\eta,\varepsilon}(Q) \in \mathcal{U}_\rho$. □

Proposition 8.5 (Lipschitz constant of the gauge)**.** *On $\mathcal{U}_\rho$ the gauge is Lipschitz in the supremum norm: $\|\mathcal{A}(Q_1) - \mathcal{A}(Q_2)\|_\infty \leq L_S(\rho)\,\|Q_1 - Q_2\|_\infty$ with $L_S(\rho) = 4/\rho$.*

Proof. Let $h = Q_1 - Q_2$ and split $\mathcal{A}(Q_1) - \mathcal{A}(Q_2)$ over the common denominator $\tilde{N}(Q_2)$, isolating the numerator difference and the denominator difference $\tilde{N}(Q_1) - \tilde{N}(Q_2) = \tilde{N}(h)$. Since T_ε is row-stochastic, $\|T_\varepsilon h\|_\infty \leq \|h\|_\infty$ and $|\tilde{N}(h)| \leq 2\|h\|_\infty$; with $0 \leq \mathcal{A}(Q_1) \leq 1$ and $\tilde{N}(Q_i) \geq \rho$, the numerator-difference term contributes $\leq 2\|h\|_\infty/\rho$ (using $\|T_\varepsilon h - (T_\varepsilon h)(1)\|_\infty \leq 2\|h\|_\infty$) and the denominator-difference term $\leq 2\|h\|_\infty/\rho$, giving $\|\mathcal{A}(Q_1) - \mathcal{A}(Q_2)\|_\infty \leq 4\|h\|_\infty/\rho$. □

Theorem 8.6 (Existence and uniqueness of the monotone fixed point)**.** *Fix $\varepsilon > 0$.* (i) *For every $\eta \in (0,1)$ and gap threshold $0 < \rho \leq \eta\,\tilde{N}(Q_{\mathrm{ref}})$, $\mathcal{R}_{\eta,\varepsilon}$ has a fixed point $Q^* \in \mathcal{U}_\rho$.* (ii) *If $(1-\eta)\,L_S(\rho) < 1$, the fixed point is unique in $\mathcal{U}_\rho$ and the iteration converges geometrically in $\|\cdot\|_\infty$.* (iii) *For strictly positive inputs $f, g \gg 0$ the smoothing channel contracts the (scale-invariant) Hilbert metric, $d_H(T_\varepsilon f, T_\varepsilon g) \leq \kappa(\varepsilon)\,d_H(f,g)$ with $\kappa(\varepsilon) = \tanh(\Delta(T_\varepsilon)/4) < 1$ (for anchored inputs vanishing at an endpoint $d_H(f,g) = +\infty$, so the contraction is read on the strictly positive smoothed iterates $P_n = T_\varepsilon Q_n$ of Section 12); this governs the sharp convergence* rate*, separately from existence.*

Proof. **(i) Existence (Schauder).** $\mathcal{U}_\rho$ is closed, bounded and convex in $C([0,1])$. The smoothing T_ε is compact by the Arzelà–Ascoli theorem (its kernel is continuous on $[0,1]^2$), and $\mathcal{A}$ is continuous (Proposition 8.5); by Lemma 8.4 $\mathcal{R}_{\eta,\varepsilon}$ is therefore a compact, continuous self-map of $\mathcal{U}_\rho$, and Schauder's fixed-point theorem [10] yields a fixed point $Q^* \in \mathcal{U}_\rho$ for every $\eta \in (0,1)$.

(ii) Uniqueness and rate (Banach). Since Q_{ref} is fixed, $\mathcal{R}_{\eta,\varepsilon}(Q_1) - \mathcal{R}_{\eta,\varepsilon}(Q_2) = (1-\eta)(\mathcal{A}(Q_1) - \mathcal{A}(Q_2))$, so by Proposition 8.5 $\mathcal{R}_{\eta,\varepsilon}$ is a $\|\cdot\|_\infty$-contraction with constant $(1-\eta)L_S(\rho)$; when this is < 1, Banach's theorem [2] gives the unique fixed point and geometric convergence. Existence in (i) uses no Hilbert metric, which resolves the affine/projective mismatch: the damped operator $\mathcal{R}_{\eta,\varepsilon}$ is affine, not homogeneous, so it is handled by Schauder/Banach in the sup-norm, not by Birkhoff contraction.

(iii) Projective rate. By Lemma 8.1 the smoothing maps $\mathcal{K}\backslash\{0\}$ into the strictly positive cone, where d_H is defined; by Proposition 8.2 its projective diameter is $\Delta(T_\varepsilon) = 2/\varepsilon < \infty$, so Birkhoff's contraction theorem [4, 7, 12] applies to the *positive linear* operator T_ε with coefficient $\tanh(\Delta/4)$, and d_H is scale-invariant. This is the contraction rate of the smoothing channel, developed in Section 12; it is not used for the existence of Q^*. □

Lemma 8.7 (Nontriviality: the fixed point is non-constant)**.** *The fixed point lies in $\mathcal{U}_\rho \subset \mathcal{M}$ (Lemma 8.4), so $Q^*(0) = 1 \neq 0 = Q^*(1)$; a constant profile has equal endpoints, hence Q^* is non-constant. The bare Perron fixed point of the row-normalized smoothing alone—the constant—is excluded by the two-end anchoring built into $\mathcal{M}$, and the admissibility gap $\tilde{N} \geq \rho > 0$ on $\mathcal{U}_\rho$ keeps the gauge $\mathcal{A}$ well defined throughout the iteration.*

Proof. By Lemma 8.4 the iterates and their limit remain in $\mathcal{U}_\rho \subset \mathcal{M}$, where $Q(0) = 1$, $Q(1) = 0$ and $\tilde{N}(Q) \geq \rho > 0$. A constant has equal endpoints (gap 0) and so lies outside $\mathcal{M}$; in particular Q^* is non-constant. □

Remark 8.8 (Scope of the single transition). That Q^* has a *unique interior* inflection is stated here at its true strength, not over-claimed. *Symmetric case (analytic):* for a mirror-symmetric kernel the central symmetry $Q^*(x) = 1 - Q^*(1-x)$ (Corollary 9.3) places the inflection of Definition 8.11 at $\hat{a} = 1/2$. *General case (certified, not general-analytic):* for asymmetric kernels the existence and non-vanishing first-order displacement of the inflection off 1/2 is the *certified* statement of Theorem 9.4 (verified interval arithmetic), and its self-similar candidate value is Conjecture 9.8. A general analytic proof that $-(Q^*)'$ is single-peaked for *all* admissible kernels is not claimed: the TP2 variation-diminishing property [16] bounds the sign changes of the *smoothed* profile, but the Hilbert-metric limit does not by itself control the curvature of the fixed point.

Theorem 8.9 (Sharp exponential decay of the global contraction)**.** *For the Gaussian kernel, $\Delta(T_\varepsilon) = 2/\varepsilon$, so*

$$\kappa(\varepsilon) = \tanh\Big(\tfrac{1}{2\varepsilon}\Big) = 1 - 2e^{-1/\varepsilon} + O(e^{-2/\varepsilon}), \qquad 1 - \kappa(\varepsilon) \sim 2e^{-1/\varepsilon},$$

whereas the local diffusive spectral gap closes only polynomially, $1 - \lambda_1(\varepsilon) \sim C\varepsilon$ (for the periodic-heat comparison model; the interval channel inherits it under the named small-ε generator-convergence hypothesis of the proof).

Proof. The full positive-cone projective diameter of the Gaussian smoothing is $\Delta(T_\varepsilon) = 2/\varepsilon$ (Proposition 8.2; an upper bound on the monotone cone, possibly not attained), so by Theorem 8.6 $\kappa(\varepsilon) = \tanh(1/(2\varepsilon))$ is a worst-case rate. Expanding $\tanh(z) = \frac{1-e^{-2z}}{1+e^{-2z}} = 1 - 2e^{-2z} + O(e^{-4z})$ at $z = 1/(2\varepsilon)$ gives $\kappa(\varepsilon) = 1 - 2e^{-1/\varepsilon} + O(e^{-2/\varepsilon})$. The periodic heat semigroup, taken as a comparison model for the local diffusive relaxation (the row-normalized interval kernel agrees with it to leading order in ε away from the boundary), has eigenvalues $e^{-4\pi^2 n^2 \varepsilon}$, so this comparison model's local gap is $1 - e^{-4\pi^2\varepsilon} \sim 4\pi^2\varepsilon$, polynomial in ε (the actual row-normalized interval channel inherits a polynomial local gap under the named small-ε generator-convergence hypothesis $(T_\varepsilon - I)/\varepsilon \to c\mathcal{L}$, with $\mathcal{L}$ a reflecting/Neumann boundary generator of positive first spectral gap and a boundary-dependent constant). A change of scale convention rescales ε but leaves the qualitative contrast intact: global projective synchronization is exponentially harder than local smoothing. $\square$

Theorem 8.10 (Critical cooling law)**.** *For the annealed smoothing chain $P_{n+1} = T_{\varepsilon_n} P_n$ inside the renormalization (the projective-synchronization budget of Section 12; the anchored iterates Q_n converge separately in sup-norm, Theorem 8.6(ii)) with $\varepsilon_n = c/\ln n$, the cumulative contraction budget diverges iff $c \geq 1$:*

$$\sum_{n \geq 2} (1 - \kappa(\varepsilon_n)) = \infty \iff c \geq 1.$$

Proof. By Theorem 8.9, $1 - \kappa(\varepsilon_n) \sim 2e^{-1/\varepsilon_n} = 2e^{-\ln n/c} = 2\,n^{-1/c}$. The series $\sum_n n^{-1/c}$ diverges iff the exponent $1/c \leq 1$, i.e. $c \geq 1$ (p-series criterion). $\square$

Definition 8.11 (Inflection order parameter)**.** The inflection of the fixed point is $\hat{a} := \arg\max_{x \in (0,1)} \big(-(Q^*)'(x) \big)$ (well defined wherever this maximizer is unique; see Remark 8.8), the location of maximal descent of the survival profile Q^* — equivalently the peak of its density $-(Q^*)'$, and the location of maximal slope of the complementary CDF $1 - Q^*$, the same point: the operator realization of the transition order parameter of Axiom A3.

Aggregation modes. Under loading, the micro-imperfection cascade activates in one of four modes, which fix the form class of Q^*: (i) parallel-instantaneous $\to$ logistic (central-limit aggregation); (ii) serial-cascaded $\to$ Weibull (weakest-link order statistics); (iii) multiaxial $\to$ hybrid (Hill/Gompertz); (iv) continuous-accumulating $\to$ Gaussian/logistic (diffusive averaging). A substrate is in general a convex combination on the simplex $\boldsymbol{\pi} = (\pi_1, \pi_2, \pi_3, \pi_4) \in \Delta_4$; the mode identities are the basis families of the form-class vector space, and neither Theorem 8.6 nor the No-Go theorem below is affected by the mixture: only the inflection location and the mixture vector carry the substrate-specific asymmetry. The classical ductile/brittle dichotomy is the binary projection of this finer partition.

9 The No-Go Theorem

The central negative result: the inflection location, although well defined for each fixed point, is *not* a universal constant across the operator class. The mechanism is reparametrization invariance—a monotone coordinate change displaces any interior location—so the result is elementary as analysis; its force is *protective*: it forecloses the false reading of $\hat{a}$ as a natural constant and makes substrate-specific asymmetry the measured quantity rather than a defect.

Definition 9.1 (Reparametrization action and equivariant transition functional)**.** The increasing homeomorphisms $h : [0,1] \to [0,1]$ act on profiles by $(h \cdot Q)(x) := Q(h(x))$, with $h^{-1} \cdot (h \cdot Q) = Q$. A class $\mathcal{G}$ of renormalization operators (TP2, strictly positive, anchored, damped) is *reparametrization-closed* if $R \in \mathcal{G}$ implies the conjugate $R^{(h)} := h^{-1} \circ R \circ h \in \mathcal{G}$ for every such h. A transition-location functional $\tau : \mathcal{M} \to (0,1)$ is *equivariant* if $\tau(h \cdot Q) = h^{-1}(\tau(Q))$.

Theorem 9.2 (No-Go: no universal inflection constant)**.** *Let $\mathcal{G}$ be reparametrization-closed and τ equivariant. Then there is no constant $c \in (0,1)$ with $\tau(Q^*) = c$ for the fixed points Q^* of all $R \in \mathcal{G}$.*

Proof. Suppose such a c exists. Fix $R \in \mathcal{G}$ with fixed point Q^*, so $\tau(Q^*) = c$. Let $\tilde{c} \in (0,1)$ be arbitrary and choose an increasing homeomorphism h with $h(c) = \tilde{c}$ (any smooth monotone h with $h(0) = 0, h(1) = 1, h(c) = \tilde{c}$ works). Form the conjugate $R^{(h)} = h^{-1} \circ R \circ h \in \mathcal{G}$ (reparametrization closure) and the profile $\tilde{Q}^* := h^{-1} \cdot Q^*$. Then

$$R^{(h)}(\tilde{Q}^*) = (h^{-1} \circ R \circ h)(h^{-1} \cdot Q^*) = h^{-1} \cdot R(h \cdot (h^{-1} \cdot Q^*)) = h^{-1} \cdot R(Q^*) = h^{-1} \cdot Q^* = \tilde{Q}^*,$$

so $\tilde{Q}^*$ is the fixed point of $R^{(h)}$. By equivariance, $\tau(\tilde{Q}^*) = \tau(h^{-1} \cdot Q^*) = h(\tau(Q^*)) = h(c) = \tilde{c}$. Since $\tilde{c}$ was arbitrary, the transition location ranges over all of $(0,1)$ across $\mathcal{G}$, contradicting $\tau \equiv c$. $\square$

Corollary 9.3 (Symmetry recovers the constant)**.** *A universal inflection location can hold only under a structural restriction that breaks reparametrization freedom. In particular, if the kernel of R is mirror-symmetric and the fixed point is unique (the Banach regime of Theorem 8.6(ii)), the fixed-point profile satisfies the central symmetry $Q^*(x) = 1 - Q^*(1-x)$; if moreover the density $-(Q^*)'$ has a unique maximum, its inflection (Definition 8.11) is at $1/2$.*

Proof. Let S be the central involution $(SQ)(x) := 1 - Q(1-x)$. On the two-end-anchored survival profiles ($Q(0) = 1$, $Q(1) = 0$) it maps $\mathcal{M}$ to itself: SQ is nonincreasing with $(SQ)(0) = 1 - Q(1) = 1$ and $(SQ)(1) = 1 - Q(0) = 0$ (the bare domain reflection $Q \mapsto Q(1 - \cdot)$ reverses monotonicity; the range flip restores it). A mirror-symmetric kernel makes $\mathcal{R}$ commute with S, $\mathcal{R} \circ S = S \circ \mathcal{R}$; by uniqueness of the fixed point (Theorem 8.6), $Q^* = SQ^*$, i.e. $Q^*(x) = 1 - Q^*(1-x)$. Differentiating gives $(Q^*)'(x) = (Q^*)'(1-x)$, so the density is symmetric about $1/2$ and its unique maximum, the inflection of Q^*, lies at $x = 1/2$. The restriction to the centrally symmetric subclass is exactly the loss of reparametrization freedom required by Theorem 9.2. □

Theorem 9.4 (Certified non-degeneracy under asymmetry). *For the damped Gaussian family parametrized by kernel asymmetry α, under the non-degenerate-mode hypothesis $q_0''(1/2) < 0$ for the symmetric density $q_0 := -(Q^*)'$, the inflection map $\alpha \mapsto \tau(Q^*_\alpha)$ is C^1, and its first-order response at the symmetric point does not vanish. Writing*

$$\dot{Q}(1/2) := \partial_\alpha \Xi(\alpha)\big|_{\alpha=0} \in [1.8412,\ 1.8963] \neq 0$$

for the perturbation derivative *of the symmetry-breaking functional Ξ — the first variation, at the midpoint, of the fixed-point condition under kernel asymmetry — this is a rigorous enclosure obtained by verified interval arithmetic at* 256*-bit precision (Krawczyk operator with Anselone bound and box enclosure) [14, 26]. Consequently $d\tau/d\alpha|_0 \neq 0$: the inflection shifts at first order off the symmetric point. The certified spectral bound $\rho(D\mathcal{R}) < 0.45$ guarantees local geometric convergence (nonsingularity of the certified branch) at the certified parameters; global geometric convergence is the separate sup-norm Banach statement $(1-\eta)L_S(\rho) < 1$.*

Proof. By the uniform contraction of Theorem 8.6 and the certified bound $\rho(D\mathcal{R}) < 0.45 < 1$, the linearization $I - D\mathcal{R}$ is invertible, so the implicit function theorem applied to $\mathcal{R}_\alpha(Q^*_\alpha) = Q^*_\alpha$ yields a C^1 branch $\alpha \mapsto Q^*_\alpha$, hence a C^1 map $\alpha \mapsto \tau(Q^*_\alpha)$. The first-order shift of the inflection is governed by the linearized inflection condition, whose coefficient is the perturbation derivative $\dot{Q}(1/2) = \partial_\alpha \Xi(\alpha)|_0$ (not the spatial slope of the profile); its enclosure $[1.8412, 1.8963]$ is produced by a reproducible ball-arithmetic certificate with auditable load-bearing steps; the full machine-verified enclosure is reproducible via `make verify` in the accompanying computational companion (an Arb ball-arithmetic certificate bound to the theorem by a Lean-checked interpretation bridge). Since the enclosure excludes 0, $d\tau/d\alpha|_0 \neq 0$, so no neighbourhood of the symmetric kernel shares the inflection value 1/2. (We assume the symmetric density is non-degenerate at its maximum: $q_0''(1/2) < 0$ for $q_0 := -(Q^*)'$, equivalently $(Q^*)'''(1/2) \neq 0$ — a strict maximum alone permits a flat (degenerate) one, so this is a named hypothesis, not implied. Under it the linearized inflection condition has nonzero leading coefficient and $d\tau/d\alpha|_0 = -\dot{Q}(1/2)/(Q^*)'''(1/2)$ is well defined.) □

Remark 9.5 (Scope). Theorem 9.2 is a non-existence statement and is complete as proved. Theorem 9.4 certifies one explicit asymmetric direction; the general statement that τ shifts for *every* admissible asymmetry, at all (ε, η), remains an open conjecture, but the No-Go theorem does not require it: a single asymmetric kernel at which the inflection moves already refutes universality. The transfer of the No-Go to the concrete order parameter $\hat{a}$ of Definition 8.11 is Lemma 9.6 below: because $\hat{a}$ is *not* equivariant under reparametrization (unlike a quantile such as the median), it requires a direct realization argument rather than an instance of Theorem 9.2.

Lemma 9.6 (Realization of the inflection; No-Go for the concrete $\hat{a}$). *Let Q^* be a fixed point whose density $q := -(Q^*)'$ is continuous and strictly positive on* $(0, 1)$*. For every $x_0 \in (0,1)$ there is an increasing homeomorphism h of* $[0, 1]$ *with $\hat{a}(h \cdot Q^*) = x_0$. Consequently $\{\hat{a}(h \cdot Q^*) :$ h an increasing homeomorphism$\} = (0, 1)$, and no constant value of $\hat{a}$ is shared by the fixed points of all operators in a reparametrization-closed class.*

Proof. Let $F := 1 - Q^*$ be the cumulative distribution function, so $F' = q > 0$ and F is an increasing homeomorphism of $[0, 1]$. Pick any continuous density $\psi > 0$ on $(0, 1)$ with a unique maximum at x_0, and let G be its CDF. Set $h := F^{-1} \circ G$; as a composition of increasing homeomorphisms it is an increasing homeomorphism of $[0,1]$ with $h(0) = 0$ and $h(1) = 1$. By the chain rule $h'(x) = G'(x)/F'(F^{-1}(G(x))) = \psi(x)/q(h(x))$, so the density of the conjugate profile is

$$-(h \cdot Q^*)'(x) = -(Q^*)'(h(x))\, h'(x) = q(h(x))\, h'(x) = \psi(x).$$

Hence $\hat{a}(h \cdot Q^*) = \arg\max_x \psi(x) = x_0$. As x_0 ranges over $(0,1)$ the orbit identity follows; by reparametrization closure (Definition 9.1) each $h \cdot Q^*$ is the fixed point of a conjugate operator in the class, so no value of $\hat{a}$ is universal. □

Remark 9.7. The functional $\hat{a} = \arg\max_x \big(-(Q^*)'(x)\big)$ is *not* equivariant in the sense of Definition 9.1: the Jacobian factor h' in $-(Q^*)'(h(x))\, h'(x)$ displaces the mode, so $\hat{a}(h \cdot Q^*) \neq h^{-1}(\hat{a}(Q^*))$ in general — unlike an equivariant quantile such as the median $(Q^*)^{-1}(\frac{1}{2})$. Theorem 9.2 therefore does not apply to $\hat{a}$ verbatim; Lemma 9.6 supplies the conclusion directly.

Conjecture 9.8 (The self-similar candidate Σ_c)**.** A minimal self-similarity condition, whole-to-core as core-to-remainder, selects a candidate inflection: $\frac{1}{\Sigma} = \frac{\Sigma}{1-\Sigma} \iff \Sigma^2 + \Sigma - 1 = 0$, with positive root $\Sigma_c = \varphi - 1 = 1/\varphi \approx 0.618$. Three independent routes (the self-similarity condition, the damage fixed-point iteration $\Sigma_{t+1} = 1 - \Sigma_t^2$, and a dimensional optimization) converge on this root. This is a conjecture, not a theorem: by Theorem 9.2 no such value can be universal across $\mathcal{G}$, and the empirically observed inflections scatter in a structured, substrate-specific way around it. Proximity of an individual measured inflection to $1/\varphi$ is a datum, not support for universality.

10 The Second No-Go: No Universal Static Classifier under Interventional Monitoring

Theorem 9.2 eliminates universal *numerical* structure: no inflection constant survives reparametrization freedom. This section eliminates, at the deployment level of Section 4, universal *verdict* structure: no static binary classifier survives the closure of the measurement–action loop, once the published score enters the subsequent dynamics and the reception is adversive. The two results are projections of one limit picture: what is universal is the structural grammar and the monitoring architecture, never a constant and never a static verdict.

Definition 10.1 (Monitoring modes)**.** Let (x_t) be the monitored trajectory and $\hat{a}_t$ the measured order parameter (Definition 4.1). Monitoring is *observational* if the dynamics do not depend on the measurement,

$$x_{t+1} = H_0(x_t),$$

and *interventional* if the published verdict enters the update,

$$x_{t+1} = H_I(x_t, \Phi(\hat{a}_t)),$$

for a classifier Φ with values in $\{0,1\}$ ($1 =$ UNSTABLE). In materials-testing terms the observational mode is non-destructive testing, the specimen is invariant under the measurement, while the interventional mode changes the boundary conditions of the loaded system through the act of reporting.

Definition 10.2 (Self-defeating reception)**.** A system a_0 in a class $\mathcal{A}$ has *self-defeating reception* if its realized stability outcome negates every published verdict: writing $C(a_0) \in \{0,1\}$ for the outcome under interventional monitoring,

$$C(a_0) = 1 - \Phi(a_0) \qquad \text{for every static } \Phi.$$

The verdict UNSTABLE induces correction (the system stabilizes); the verdict STABLE induces complacency (the system destabilizes). This is the self-defeating prophecy [21], recorded here as a class property, not constructed from the axioms. We write

$$S_\Delta := \{a \in \mathcal{A} : \ a \text{ has self-defeating reception}\} \subseteq \mathcal{A}$$

for the subset of such systems.

Theorem 10.3 (Diagonal feedback barrier)**.** *Let $\mathcal{A}$ be a system class with $S_\Delta \neq \emptyset$. Then no static classifier $\Phi : \mathcal{A} \to \{0,1\}$ is correct on $\mathcal{A}$ under interventional monitoring.*

Proof. Correctness requires $\Phi(a) = C(a)$ for all $a \in \mathcal{A}$. For $a_0 \in S_\Delta$ this reads $\Phi(a_0) = 1 - \Phi(a_0)$, i.e. a fixed point of the negation on $\{0,1\}$; the negation has none. □

Remark 10.4 (Status of the theorem)*.* Theorem 10.3 is the terminal step of the classical diagonal argument [18, 33]: fixed-point freeness of negation. It is not new mathematics, and the negation is *defined into* the class by Definition 10.2, not constructed. Its content is locative, parallel to Theorem 9.2: it fixes *where* the impossibility lives — in the closed, adversive measurement–action loop — and thereby what monitoring may still claim there. Whether a given real class (conversational trajectories under visible scoring among them) actually has $S_\Delta \neq \emptyset$ is an empirical premise: it can be supported by a controlled mode switch $H_0 \leftrightarrow H_I$, never settled formally.

Proposition 10.5 (A closed loop is necessary, not sufficient)**.** *Let the reception of the published verdict be $\psi : \{0,1\} \to \{0,1\}$, so that the realized outcome at the loop is $C = \psi \circ \Phi$. Among the four self-maps of $\{0,1\}$, the negation is the only fixed-point-free one. Consequently: if $\psi \neq \neg$ (in particular $\psi = \mathrm{id}$, the self-fulfilling reception, and the constant receptions), then ψ has a fixed point b^* and the constant classifier $\Phi \equiv b^*$ is correct at the loop: a regime-bound classifier exists. Only the adversive loop $\psi = \neg$ raises the barrier.*

Proof. id fixes both points, the constants fix their value, and $\neg$ exchanges the two points. If $\psi(b^*) = b^*$, publishing b^* yields outcome $\psi(b^*) = b^*$, which is the published verdict. □

Remark 10.6 (The $\Phi_\mathcal{R}$ diagnosis: two edges of one fixed-point picture)*.* The existence question for a correct classifier is in both classical traditions a fixed-point question for the reaction map, answered at two different edges. On the continuous edge, a public prediction with *continuous* reaction admits a correct value by the intermediate-value/Brouwer argument, the classical possibility theorem of public prediction [13, 30]. On the binary edge, Proposition 10.5 locates the unique exception: only the adversive reception is fixed-point free. Continuity buys the fixed point; discreteness localizes its only failure.

The tempting strengthening, "a regime-bound classifier exists *iff* no reception coordinate is the negation", holds only under *separability* (memoryless receptions acting coordinatewise, $\psi = (r_a)_a$). It fails in the coupled case: on $\{0,1\}^2$ the rotation $\psi(x,y) = (y, 1-x)$ is fixed-point free although neither coordinate negates itself: of the nine fixed-point-free permutations of the four states, only three are separable. Adversiveness can be *distributed* across coupled axes (x follows y, y opposes x) without being attributable to any single axis. Since specimens carry loading-path memory — $\hat{a}_t$ is a functional of the whole history $x_{0:t}$ (Axioms A2, A4), and the deformation axes are coupled (Remark 4.2) — the non-separable case is the generic one for this framework. The characterization therefore remains a *diagnosis*, not a theorem: it unifies the two edges and names the exact residue (the sign of the coupled loop), which is an empirical question, not a structural one.

The design consequence is the operational counterpart of Theorem 9.2: monitoring must be specified as $\Phi_\mathcal{R}$ — bound to an explicit observation, feedback and intervention regime $\mathcal{R}$ — and never as a universal Φ_{univ}. This is the architecture already adopted in Definition 4.3: thresholds are calibration parameters of a regime, the state machine is the invariant. The continuous-optimization literature on prediction that shifts its own distribution [23] studies retraining equilibria for risk minimizers; the binary verdict cut and its barrier are a separate, elementary stratum beneath it.

Remark 10.7 (Two No-Gos, one picture). Theorem 9.2 forbids a universal *value* by reparametrization freedom of the substrate; Theorem 10.3 forbids a universal *static verdict* by fixed-point freeness of the adversive loop. Both eliminate universals in favour of structure: the functional form, the accumulation architecture, and the regime-bound measurement survive; constants and context-free classifications do not. The first No-Go makes the inflection a measured quantity; the second makes the verdict a regime property. Together they state the honest scope of the framework's operational claim: structural grammar universal, numbers and verdicts local.

11 The Elimination Ledger

We collect, in reader-checkable form, what *cannot* be concluded at the present level of generality, and the minimal structure that would change each conclusion. These are not statements of empirical uncertainty: each is a consequence of the preceding theorems.

Proposition 11.1 (E1: no universal rate from local contractivity). *Per-step contractivity (existence of $\kappa_n < 1$) forces projective synchronization (loss of initialization) of the smoothing chain once the budget diverges (Theorem 8.10), anchored convergence to Q^* being the separate sup-norm Banach statement, but does not determine a convergence-rate class (exponential / polynomial / ...) across systems: the rate is set by the cumulative budget — exactly the projective budget $B_n := -\sum_{k\le n}\log\kappa_k$, with linearization $A_n := \sum_{k\le n}(1-\kappa_k) \le B_n$ (equal as $\kappa_n \to 1$) — not by any single κ_n.*

Proof. By Theorem 8.6(iii) each smoothing contracts d_H by κ_k, so two smoothed trajectories synchronize at the accumulated rate, $d_H(P_n, P'_n) \le \left(\prod_{k\le n}\kappa_k\right) d_H(P_0, P'_0) \le e^{-A_n} d_H(P_0, P'_0)$, using $\kappa_k \le e^{-(1-\kappa_k)}$. The bound depends on A_n only; the example $\kappa_n = \tanh(1/(2\varepsilon_n))$ with $\varepsilon_n = c/\ln n$ (Theorem 8.10) gives $A_n \sim \sum n^{-1/c}$, whose growth class (and hence the rate) is tuned continuously by c while every local $\kappa_n < 1$. ☐

Proposition 11.2 (E2: sharpness of budget bounds). *Even with an a-priori bound $d_n \le e^{-A_n} d_0$, no universal improvement holds within the class: the worst-case Birkhoff budget can be scheduled arbitrarily slowly — for any prescribed decay profile there is an admissible schedule (ε_n) realizing that budget (on the monotone anchored subclass the realized Hilbert diameter may be strictly smaller than the full-cone bound).*

Proof. Given a target nonincreasing profile $\beta_n \downarrow 0$, choose κ_n with $\prod_{k\le n}\kappa_k = \beta_n$ (i.e. $1-\kappa_n = 1-\beta_n/\beta_{n-1}$), realizable by $\varepsilon_n = 1/(2\,\mathrm{artanh}\,\kappa_n)$ since $\kappa \mapsto \tanh(1/(2\varepsilon))$ is a bijection onto $(0,1)$. Each $\kappa_n < 1$ is admissible, so the decay can be made arbitrarily slow. ☐

Proposition 11.3 (E3: schedule dependence). *When the per-step contraction degrades ($\kappa_n \to 1$, e.g. $\varepsilon_n \to 0$), convergence hinges on the explicit divergence $\sum_n(1-\kappa_n) = \infty$ (Theorem 8.10); absent this schedule condition, persistent bad windows or non-convergence cannot be excluded.*

Proposition 11.4 (E4: no universal numerical threshold). *No concrete inflection/transition value (e.g. $1/2$, $1/\varphi$) is derivable at this level of generality; by Theorem 9.2 for equivariant transition locations, and by Lemma 9.6 for the non-equivariant modal inflection $\hat{a}$, any such value requires structure that breaks reparametrization freedom.*

Proposition 11.5 (E5: no universal static classifier)**.** *Correctness of a static classifier* $\Phi : \mathcal{A} \to \{0,1\}$ *on a feedback-responsive class cannot be concluded without an empirically established reception sign for the deployed regime: if* $S_\Delta \neq \emptyset$*, no such* Φ *exists (Theorem 10.3); only* $\Phi_{\mathcal{R}}$*-statements relative to a specified observation–feedback–intervention regime are admissible.*

Minimal structure that changes the conclusions.

(S1) Uniform minorization (Doeblin).
A genuine lower bound $Tf \geq c\,\langle f, \nu\rangle$ yields uniform mixing and a uniform rate (Section 12); converts E1/E3 into an explicit exponential rate.

(S2) Time-structure (good-step density / bounded gaps).
A lower bound on the frequency of contractive steps converts local contractivity into an explicit rate in terms of that density; addresses E1/E2.

(S3) Symmetry / identifiability.
Restriction to a mirror-symmetric or identifiable normal-form subclass pins a numerical inflection location (Corollary 9.3); addresses E4.

12 Spectral and Contraction Substructure

The rate of approach to the autopoietic fixed point is governed by the contraction budget, not by any local coefficient. We make the governing structure precise and exhibit the uniform-rate restoration (S1). Throughout, the rate theory is for the *linear smoothing chain* $P_{n+1} := T_{\varepsilon_n} P_n$ on strictly positive profiles (Lemma 8.1), where the Hilbert metric is defined and the Birkhoff contraction of T_ε applies directly (Theorem 8.6(iii)); this is the channel's *global projective synchronization.* The anchored damped iterates $Q_n = \mathcal{R}_{\eta,\varepsilon}(Q_{n-1})$ vanish at the right endpoint (so are not compared in d_H, and $\mathcal{R}$ being affine is not Birkhoff-contractive); their convergence $Q_n \to Q^*$ is the separate sup-norm Banach statement of Theorem 8.6(ii), *not* inferred from Hilbert closeness of the smoothed profiles.

Definition 12.1 (Contraction budget)**.** For a (possibly non-autonomous) iteration with per-step coefficients $\kappa_n \in (0,1)$, the cumulative contraction budget is $A_n := \sum_{k\leq n}(1-\kappa_k)$.

Theorem 12.2 (Budget governs convergence)**.** *For the smoothing channel, two smoothed trajectories* P_n, P_n' *synchronize at the budget rate,* $d_H(P_n, P_n') \leq e^{-A_n} d_H(P_0, P_0')$*; in particular* $A_n \to \infty$ *forces global synchronization (*$d_H(P_n, P_n') \to 0$*), while a bounded budget leaves a non-vanishing contraction factor and so cannot force it. Convergence of the anchored iterates* $Q_n \to Q^*$ *is the sup-norm statement of Theorem 8.6(ii). The exact projective rate is governed by* $B_n := -\sum_{k<n}\log\kappa_k$ *(so* $d_H(P_n, P_n') \leq e^{-B_n} d_H(P_0, P_0')$*,* $B_n \geq A_n$*);* A_n *is its linearization, controlling the divergence dichotomy and coinciding with* B_n *in the cooling regime* $\kappa_n \to 1$*. The asymptotic rate class (the growth class of* B_n*, equivalently of* A_n *as* $\kappa_n \to 1$*) is invariant under any reparametrization that preserves it.*

Proof. The product bound $\prod_{k\leq n}\kappa_k \leq e^{-\sum_{k\leq n}(1-\kappa_k)} = e^{-A_n}$ gives the estimate (each smoothing contracts d_H by κ_k, Theorem 8.6(iii)); if $A_n \to \infty$ then $d_H(P_n, P_n') \to 0$. Conversely, if A_n is bounded then $\kappa_k \to 1$, so $\kappa_k \geq e^{-2(1-\kappa_k)}$ holds for $k \geq N$ (some finite N), and $\prod_{k\leq n}\kappa_k \geq \big(\prod_{k<N}\kappa_k\big)\exp\big(-2\sum_{k\geq N}(1-\kappa_k)\big) \geq \big(\prod_{k<N}\kappa_k\big)e^{-2A_\infty} > 0$, so the contraction factor does not vanish and convergence to a unique limit cannot be forced. Reparametrization-invariance of the rate follows because A_n is built from the projective coefficients κ_k, which are gauge-invariant (Theorem 8.6). $\square$

Theorem 12.3 (Doeblin restart: uniform rate (S1))**.** *Replace T_ε by the regularized kernel $k_{\zeta,\varepsilon} = (1-\zeta)k_\varepsilon + \zeta$ in the rate theory (a uniform minorization of weight ζ, a* kernel-level *floor distinct from the operator damping η of Definition 8.3; the floor secures positivity for the Birkhoff estimate but need not preserve the TP2/monotone structure, so it is applied to the smoothing chain P_n, not to the anchored gauge). Then the projective contraction satisfies $\kappa_\zeta \le \frac{1-\zeta}{1+\zeta} < 1$ uniformly in ε, and: **(i)** loss of initialization at the budget rate whenever $\sum_n \zeta_n = \infty$ — two smoothed trajectories satisfy $d_H(P_n, P'_n) \le e^{-A_n} d_H(P_0, P'_0)$; and **(ii)** for the anchored iteration with the pure TP2 kernel k_ε, under a uniform admissibility gap $0 < \rho \le \eta \inf_n \tilde{N}_{\varepsilon_n}(Q_{\mathrm{ref}})$ with $(1-\eta)L_S(\rho) < 1$ (so every iterate lies in one complete class $\mathcal{U}_\rho$ with a single Banach constant), convergence $Q_n \to Q_0^*$ to the limiting fixed point,* provided *the sup-norm fixed-point drift $\Delta_n := \left\| Q^*_{\varepsilon_{n-1}} - Q^*_{\varepsilon_n} \right\|_\infty \to 0$ and $Q^*_{\varepsilon_n} \to Q^*_0$ in sup-norm.*

Proof. A uniform additive component ζ is a Doeblin minorization $k_{\zeta,\varepsilon} \ge \zeta$; by the Doeblin coupling bound [11, 29] the Hilbert (Birkhoff) contraction is at most $\frac{1-\zeta}{1+\zeta}$, from the bounded kernel ratio $k_{\zeta,\varepsilon} \in [\zeta, 1]$, independent of ε (the total-variation Doeblin coefficient is the looser $1-\zeta$). **(i)** Then $1 - \kappa_{\zeta_n} \ge \zeta_n$, so $A_n \ge \sum_{k \le n} \zeta_k$, and each step contracts pairwise (the minorized kernel is strictly positive, Birkhoff): $d_H(P_n, P'_n) \le \prod_{k<n} \kappa_{\zeta_k} \, d_H(P_0, P'_0) \le e^{-A_n} d_H(P_0, P'_0)$, no common fixed point required. **(ii)** The anchored iteration retains the pure TP2 kernel k_ε, so Lemma 8.4's monotone self-map is intact (the additive Doeblin floor of (i) does not enter it). Convergence to the limiting fixed point is the *anchored* sup-norm statement, not a Hilbert one — $T_\varepsilon Q^*_\varepsilon$ is not a fixed point of the linear smoothing chain (whose Perron limit is the constant profile), so the rate toward Q^*_0 cannot be read off d_H. With $e_n := \|Q_n - Q^*_{\varepsilon_n}\|_\infty$, the Banach contraction $\left\| \mathcal{R}_{\eta,\varepsilon_n} Q - \mathcal{R}_{\eta,\varepsilon_n} \widetilde{Q} \right\|_\infty \le (1-\eta)L_S(\rho) \left\| Q - \widetilde{Q} \right\|_\infty$ (uniform in n, < 1 in the damping regime) toward $Q^*_{\varepsilon_{n-1}}$, followed by the fixed-point displacement $\Delta_n := \left\| Q^*_{\varepsilon_{n-1}} - Q^*_{\varepsilon_n} \right\|_\infty$, gives $e_n \le (1-\eta)L_S(\rho)\, e_{n-1} + \Delta_n$; a constant contraction < 1 with vanishing forcing $\Delta_n \to 0$ forces $e_n \to 0$, i.e. $Q_n \to Q^*_0$ in sup-norm. The full-annealing limit $\varepsilon_n \to 0$ is singular (the projective contraction estimate degenerates, Theorem 8.9), so (ii) is read for cooling schedules with a non-degenerate limit. $\square$

Remark 12.4 (Local versus global gap)*.* Theorem 8.9 exhibits the central discrepancy: the *local* diffusive relaxation gap closes polynomially ($1 - \lambda_1 \sim C\varepsilon$), while the *global* projective synchronization gap closes exponentially ($1 - \kappa \sim 2e^{-1/\varepsilon}$). Local smoothing is easy; global ordering is exponentially hard. The certified spectral bound $\rho(D\mathcal{R}) < 0.45$ (Theorem 9.4) shows that at fixed, non-vanishing scale the realized linear gap is far wider than the Birkhoff-theoretic worst case: the degeneration is a singular-limit ($\varepsilon \to 0$) phenomenon, not a defect at operating scale. This is the spectral content behind the minimum-sampling requirement of the operational frame: a trajectory must accumulate enough budget for the inflection to be reconstructible.

Remark 12.5 (Closing)*.* The framework is complete in the following sense. Sections 1–2 fix the class and reduce accumulation, transition and irreversibility to self-reference and dissipation. Section 3 transports the structure across scales. Section 4 instantiates it. Sections 5–7 prove the three structural theorems. Sections 8–9 construct the order-parameter fixed point and prove that its location is substrate-specific, not universal. Section 10 extends the elimination to the deployment level: under interventional monitoring with adversive reception no universal static classifier exists, and verdicts are regime-bound. Sections 11–12 delimit the rate theory. Every numerical inflection is, by Theorem 9.2, a measured quantity; what is universal is the structural grammar: the recurrence of one functional form, read off one cumulative observable, across substrates.

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